

Project: Customer Churn Prediction

This project aimed to build a machine learning classification model to predict customer churn using the Customer Churn Prediction dataset from Kaggle. The project followed a structured data science methodology covering data assessment, exploratory data analysis, preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation.

Phase 1 Data Collection

data was collected in kaggle:

<https://www.kaggle.com/datasets/muhammadshahidazeem/customer-churn-dataset/data>

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

Phase 2 Initial Data Assessment

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
%matplotlib inline

training_path =
'/content/drive/MyDrive/KaggleDatasets/CustomerChurn/1/customer_churn_
dataset-training-master.csv'
testing_path =
'/content/drive/MyDrive/KaggleDatasets/CustomerChurn/1/customer_churn_
dataset-testing-master.csv'

train = pd.read_csv(training_path)
test = pd.read_csv(testing_path)

# checking the number of records and columns
print(f"training has {train.shape[0]} rows and {train.shape[1]}
columns")
print(f"testing has {test.shape[0]} rows and {test.shape[1]} columns")

training has 440833 rows and 12 columns
testing has 64374 rows and 12 columns

train.columns

Index(['CustomerID', 'Age', 'Gender', 'Tenure', 'Usage Frequency',
       'Support Calls', 'Payment Delay', 'Subscription Type',
```

```
'Contract Length', 'Total Spend', 'Last Interaction', 'Churn'],  
dtype='object')
```

```
# checking for training class imbalance
```

```
print(train['Churn'].value_counts(normalize=True) * 100)
```

```
Churn
```

```
1.0    56.71072
```

```
0.0    43.28928
```

```
Name: proportion, dtype: float64
```

```
sns.countplot(x='Churn', data=train)
```

```
plt.title('Training Churn Class Distribution')
```

```
Text(0.5, 1.0, 'Training Churn Class Distribution')
```



```
print(test['Churn'].value_counts(normalize=True) * 100)
```

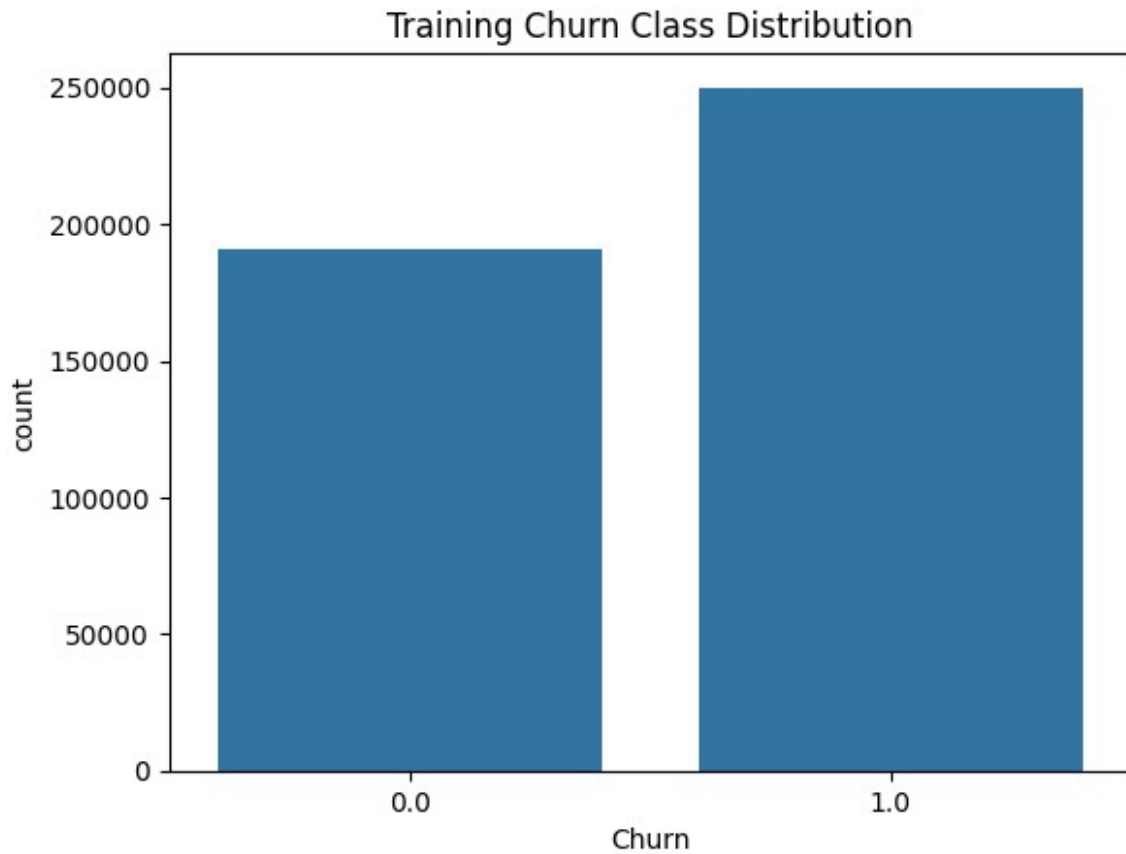
```
Churn
```

```
0    52.631497
```

```
1    47.368503
```

```
Name: proportion, dtype: float64
```

```
sns.countplot(x='Churn', data=test)
plt.title('Training Churn Class Distribution')
Text(0.5, 1.0, 'Training Churn Class Distribution')
```



```
# check the number of missing values
train.isna().sum()
```

CustomerID	1
Age	1
Gender	1
Tenure	1
Usage Frequency	1
Support Calls	1
Payment Delay	1
Subscription Type	1
Contract Length	1
Total Spend	1
Last Interaction	1
Churn	1

dtype: int64

```

train.dropna(inplace=True)
test.dropna(inplace=True)

test.isna().sum()

CustomerID      0
Age             0
Gender          0
Tenure          0
Usage Frequency 0
Support Calls    0
Payment Delay    0
Subscription Type 0
Contract Length 0
Total Spend     0
Last Interaction 0
Churn           0
dtype: int64

# checking for duplicate records
print(f"number of duplicates in training: {train.duplicated().sum()}")
print(f"number of duplicates in testing: {test.duplicated().sum()}")

number of duplicates in training: 0
number of duplicates in testing: 0

train.head()

{"type": "dataframe", "variable_name": "train"}

train.info()

<class 'pandas.core.frame.DataFrame'>
Index: 440832 entries, 0 to 440832
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   CustomerID            440832 non-null  float64
 1   Age                  440832 non-null  float64
 2   Gender               440832 non-null  object
 3   Tenure               440832 non-null  float64
 4   Usage Frequency      440832 non-null  float64
 5   Support Calls        440832 non-null  float64
 6   Payment Delay        440832 non-null  float64
 7   Subscription Type     440832 non-null  object
 8   Contract Length      440832 non-null  object
 9   Total Spend          440832 non-null  float64
10   Last Interaction     440832 non-null  float64
11   Churn                440832 non-null  float64
dtypes: float64(9), object(3)
memory usage: 43.7+ MB

```

```

columns_change_to_int = ['CustomerID', 'Tenure', 'Age', 'Usage
Frequency', 'Support Calls', 'Payment Delay', 'Last Interaction', 'Churn']
train[columns_change_to_int] =
train[columns_change_to_int].astype(int)
columns_change_to_categorical = ['Gender', 'Subscription
Type', 'Contract Length']
train[columns_change_to_categorical] =
train[columns_change_to_categorical].astype('category')

train.info()

```

```

<class 'pandas.core.frame.DataFrame'>
Index: 440832 entries, 0 to 440832
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            440832 non-null  int64
1   Age                   440832 non-null  int64
2   Gender                440832 non-null  category
3   Tenure                440832 non-null  int64
4   Usage Frequency       440832 non-null  int64
5   Support Calls         440832 non-null  int64
6   Payment Delay         440832 non-null  int64
7   Subscription Type     440832 non-null  category
8   Contract Length       440832 non-null  category
9   Total Spend           440832 non-null  float64
10  Last Interaction      440832 non-null  int64
11  Churn                 440832 non-null  int64
dtypes: category(3), float64(1), int64(8)
memory usage: 34.9 MB

```

```

test.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 64374 entries, 0 to 64373
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   CustomerID            64374 non-null  int64
1   Age                   64374 non-null  int64
2   Gender                64374 non-null  object
3   Tenure                64374 non-null  int64
4   Usage Frequency       64374 non-null  int64
5   Support Calls         64374 non-null  int64
6   Payment Delay         64374 non-null  int64
7   Subscription Type     64374 non-null  object
8   Contract Length       64374 non-null  object
9   Total Spend           64374 non-null  int64
10  Last Interaction      64374 non-null  int64
11  Churn                 64374 non-null  int64

```

```

dtypes: int64(9), object(3)
memory usage: 5.9+ MB

test['Total Spend'] = test['Total Spend'].astype(float)
test['Churn'] = test['Churn'].astype(int)
columns_change_to_categorical = ['Gender', 'Subscription
Type', 'Contract Length']
test[columns_change_to_categorical] =
test[columns_change_to_categorical].astype('category')

```

Phase 3: Exploratory Data Analysis and Data Cleaning

```

train.drop('CustomerID', axis=1, inplace=True)
test.drop('CustomerID', axis=1, inplace=True)

train.head()

{"type": "dataframe", "variable_name": "train"}

test.head()

{"summary": "{\n  \"name\": \"test\",\n  \"rows\": 64374,\n  \"fields\": [\n    {\n      \"column\": \"Age\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 13,\n        \"min\": 18,\n        \"max\": 65,\n        \"num_unique_values\": 48,\n        \"samples\": [\n          57,\n          20,\n          61\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Gender\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 2,\n        \"samples\": [\n          \"Male\",\n          \"Female\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Tenure\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 17,\n        \"min\": 1,\n        \"max\": 60,\n        \"num_unique_values\": 60,\n        \"samples\": [\n          25,\n          41\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Usage Frequency\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 8,\n        \"min\": 1,\n        \"max\": 30,\n        \"num_unique_values\": 30,\n        \"samples\": [\n          9,\n          2\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Support Calls\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 3,\n        \"min\": 0,\n        \"max\": 10,\n        \"num_unique_values\": 11,\n        \"samples\": [\n          10,\n          4\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"Payment Delay\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 8,\n        \"min\": 0,\n        \"max\": 10,\n        \"num_unique_values\": 11,\n        \"samples\": [\n          10,\n          4\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      }\n    ]\n  }\n}"}

```

```

\ "max\ ": 30,\n          \ "num_unique_values\ ": 31,\n          \ "samples\ ":
[\n          26,\n          15\n          ],\n          \ "semantic_type\ ":
\ "\",\n          \ "description\ ": \ "\",\n          }\n          },\n          {\n
\ "column\ ": \ "Subscription Type\ ",\n          \ "properties\ ": {\n
\ "dtype\ ": \ "category\ ",\n          \ "num_unique_values\ ": 3,\n
\ "samples\ ": [\n          \ "Basic\ ",\n          \ "Standard\ "
n          ],\n          \ "semantic_type\ ": \ "\",\n
\ "description\ ": \ "\",\n          }\n          },\n          {\n          \ "column\ ":
\ "Contract Length\ ",\n          \ "properties\ ": {\n          \ "dtype\ ":
\ "category\ ",\n          \ "num_unique_values\ ": 3,\n          \ "samples\ ":
[\n          \ "Monthly\ ",\n          \ "Annual\ "\n          ],\n
\ "semantic_type\ ": \ "\",\n          \ "description\ ": \ "\",\n          }\n
n          },\n          {\n          \ "column\ ": \ "Total Spend\ ",\n
\ "properties\ ": {\n          \ "dtype\ ": \ "number\ ",\n          \ "std\ ":
260.8748090082538,\n          \ "min\ ": 100.0,\n          \ "max\ ": 1000.0,\n
n          \ "num_unique_values\ ": 901,\n          \ "samples\ ": [\n
867.0,\n          278.0\n          ],\n          \ "semantic_type\ ": \ "\",\n
n          \ "description\ ": \ "\",\n          }\n          },\n          {\n
\ "column\ ": \ "Last Interaction\ ",\n          \ "properties\ ": {\n
\ "dtype\ ": \ "number\ ",\n          \ "std\ ": 8,\n          \ "min\ ": 1,\n
\ "max\ ": 30,\n          \ "num_unique_values\ ": 30,\n          \ "samples\ ":
[\n          7,\n          22\n          ],\n          \ "semantic_type\ ":
\ "\",\n          \ "description\ ": \ "\",\n          }\n          },\n          {\n
\ "column\ ": \ "Churn\ ",\n          \ "properties\ ": {\n          \ "dtype\ ":
\ "number\ ",\n          \ "std\ ": 0,\n          \ "min\ ": 0,\n
\ "max\ ": 1,\n          \ "num_unique_values\ ": 2,\n          \ "samples\ ":
[\n          0,\n          1\n          ],\n          \ "semantic_type\ ":
\ "\",\n          \ "description\ ": \ "\",\n          }\n          }\n          ]\n
n}", "type": "dataframe", "variable_name": "test"}

```

```

from scipy import stats
from scipy.stats import pointbiserialr

def cramers_v(x, y):
    ''' function to see the association between two categorical
    data'''
    confusion_matrix = pd.crosstab(x, y)
    chi2 = stats.chi2_contingency(confusion_matrix)[0]
    n = confusion_matrix.sum().sum()
    phi2 = chi2 / n
    r, k = confusion_matrix.shape
    return np.sqrt(phi2 / min(k-1, r-1))

def eta_squared(numeric_col, cat_col):
    ''' function to see the association between numeric column and
    categorical column with 2+ categories'''
    groups = [numeric_col[cat_col == c] for c in cat_col.unique()]
    return stats.f_oneway(*groups)[0] # F-statistic as proxy

def association_matrix(df, num_cols, cat_cols):

```

```

''' function to see the association between all column of the
dataset, used to check for multi colinearity'''
all_cols = num_cols + cat_cols
matrix = pd.DataFrame(np.ones((len(all_cols), len(all_cols))),
                      index=all_cols, columns=all_cols)
for i, c1 in enumerate(all_cols):
    for j, c2 in enumerate(all_cols):
        if i == j:
            matrix.loc[c1, c2] = 1.0
        elif c1 in num_cols and c2 in num_cols:
            matrix.loc[c1, c2] = df[c1].corr(df[c2],
method='spearman')
        elif c1 in cat_cols and c2 in cat_cols:
            matrix.loc[c1, c2] = cramers_v(df[c1], df[c2])
        else:
            num = c1 if c1 in num_cols else c2
            cat = c1 if c1 in cat_cols else c2
            corr, _ = pointbiserialr(df[num],
pd.factorize(df[cat])[0])
            matrix.loc[c1, c2] = abs(corr)
    return matrix

# checking the association of the feature columns to the target column
num_cols = ['Tenure', 'Age', 'Usage Frequency', 'Support Calls',
'Payment Delay', 'Last Interaction', 'Total Spend']
cat_cols = ['Gender', 'Subscription Type', 'Contract Length', 'Churn']

for col in num_cols:
    # since we are comparing numerical column to the target (which is
churn that is a categorical column with 2 categories) we will use
point biserialr to know its correlation
    corr, pval = pointbiserialr(train[col], train['Churn'])
    print(f"{col}: r={corr:.3f}, p={pval:.4f}")

for col in cat_cols:
    # since we have categorical columns in the dataset we will call
the function the cremers to know the association
    print(f"{col}: V={cramers_v(train[col], train['Churn']):.3f}")

Tenure: r=-0.052, p=0.0000
Age: r=0.218, p=0.0000
Usage Frequency: r=-0.046, p=0.0000
Support Calls: r=0.574, p=0.0000
Payment Delay: r=0.312, p=0.0000
Last Interaction: r=0.150, p=0.0000
Total Spend: r=-0.429, p=0.0000
Gender: V=0.175
Subscription Type: V=0.021
Contract Length: V=0.434
Churn: V=1.000

```


after checking the associations of the targets we concluded that we can remove Tenure, Usage Frequency, Subscription Type, Last Interaction, and Gender columns since it shows a low correlation with the target column

```
# dropping columns
train.drop(['Tenure', 'Usage Frequency', 'Subscription
Type', 'Gender', 'Last Interaction'], axis=1, inplace=True)
test.drop(['Tenure', 'Usage Frequency', 'Subscription
Type', 'Gender', 'Last Interaction'], axis=1, inplace=True)

# checking multicollinearity between feature columns
num_cols = ['Age', 'Support Calls', 'Payment Delay', 'Total Spend']
cat_cols = ['Contract Length']
association_matrix(train, num_cols, cat_cols)

{"summary": "{\n  \"name\": \"association_matrix(train, num_cols,\n  cat_cols)\",\n  \"rows\": 5,\n  \"fields\": [\n    {\n      \"column\": \"Age\", \n      \"properties\": {\n        \"dtype\":\n        \"number\", \n        \"std\": 0.43917867516609604, \n        \"min\": -\n        0.07035708699445974, \n        \"max\": 1.0, \n        \"num_unique_values\": 5, \n        \"samples\": [\n          0.16861721348553366, \n          0.00033353012615250236, \n          0.051442665863143604\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }, \n      \"column\": \"Support Calls\", \n      \"properties\": {\n        \"dtype\":\n        \"number\", \n        \"std\": 0.4583329479377034, \n        \"min\": -\n        0.19909465416220998, \n        \"max\": 1.0, \n        \"num_unique_values\": 5, \n        \"samples\": [\n          0.00045758169617369105, \n          0.14618123968598534\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }, \n      \"column\": \"Payment Delay\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\":\n        0.4459865404989409, \n        \"min\": -0.10406656002824327, \n        \"max\": 1.0, \n        \"num_unique_values\": 5, \n        \"samples\":\n        [\n          0.14618123968598534, \n          0.00012723100523885585, \n          1.0\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }, \n      \"column\": \"Total Spend\", \n      \"properties\": {\n        \"dtype\":\n        \"number\", \n        \"std\": 0.49415641429597185, \n        \"min\": -\n        0.19909465416220998, \n        \"max\": 1.0, \n        \"num_unique_values\": 5, \n        \"samples\": [\n          0.19909465416220998, \n          0.0006419527335179895, \n          0.10406656002824326\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\"\n      }, \n      \"column\": \"Contract Length\", \n      \"properties\": {\n        \"dtype\":\n        \"number\", \n        \"std\": 0.4470391883788024, \n        \"min\":\n        0.00012723100523885585, \n        \"max\": 1.0, \n        \"num_unique_values\": 5, \n        \"samples\": [\n          0.00045758169617369105, \n          1.0\n        ]\n      }\n    ]\n  }\n}
```

```
0.00012723100523885585\n          ],\n          \"semantic_type\": \"\", \n          \"description\": \"\" \n        }\n      }\n    ]\n  },\n  \"type\": \"dataframe\"}
```

since there are no multicollinearity we will not drop any columns now

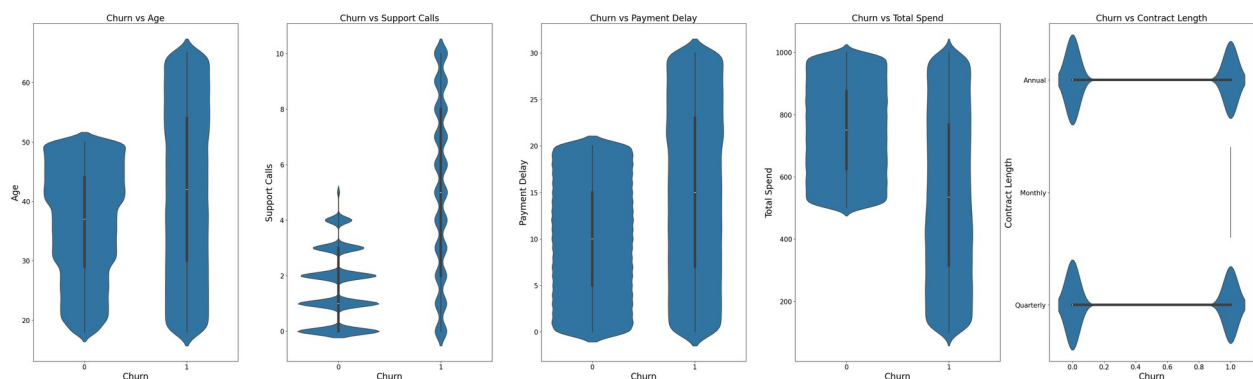
```
# checking the distributions of every feature column with respect to
the target feature
```

```
features = ['Age', 'Support Calls', 'Payment Delay', 'Total
Spend', 'Contract Length']
```

```
fig, axes = plt.subplots(1, len(features), figsize=(40, 12))
```

```
for i, feature in enumerate(features):
    sns.violinplot(data=train, x='Churn', y=feature, ax=axes[i])
    axes[i].set_title(f'Churn vs {feature}', fontsize=20)
    axes[i].set_xlabel('Churn', fontsize=20)
    axes[i].set_ylabel(feature, fontsize=20)
    axes[i].tick_params(axis='both', labelsize=15)
```

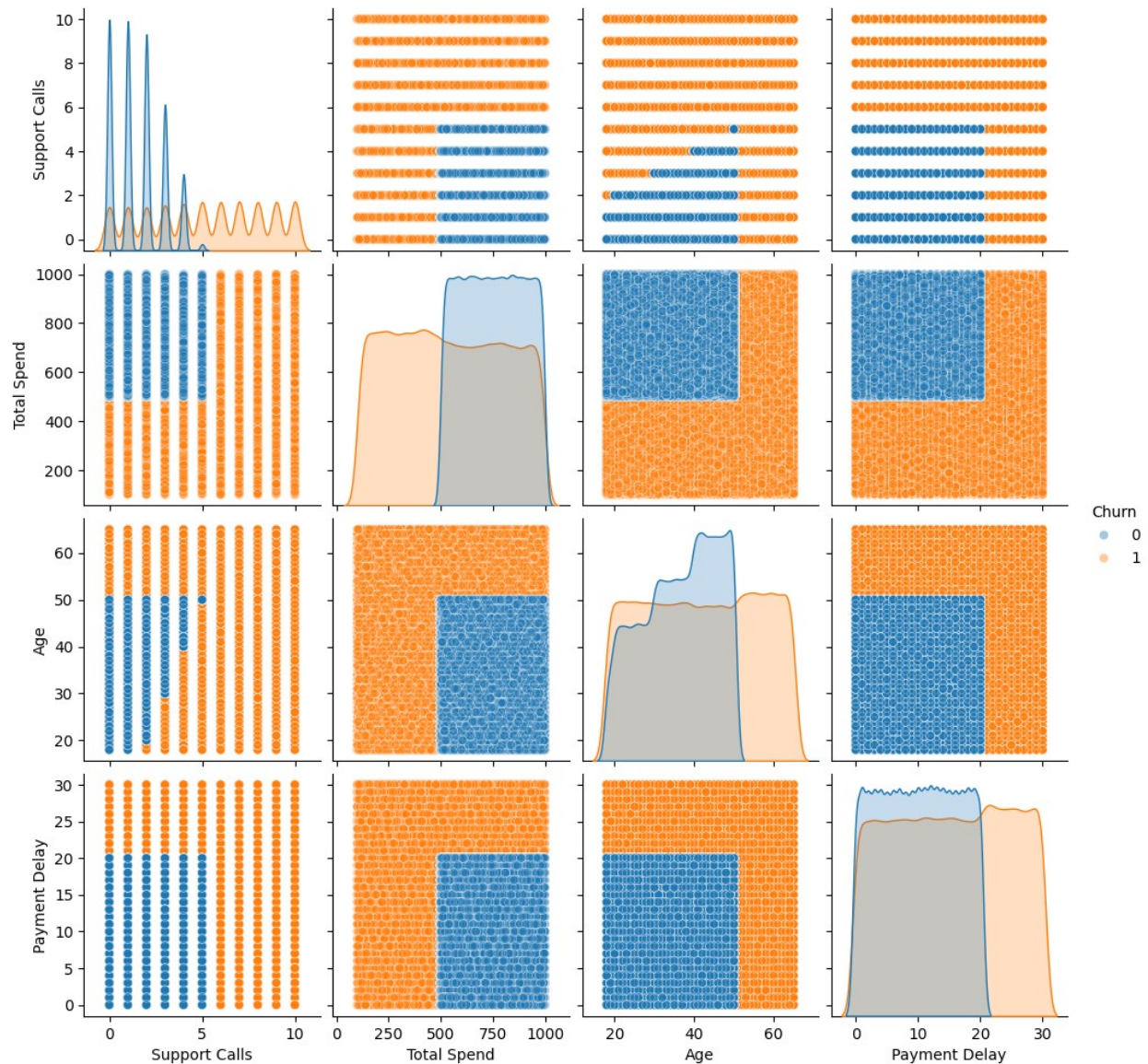
```
plt.tight_layout()
```



analysis: Churn vs Age: Older customers churn slightly more Churn vs Support Calls: customer with 5 or more support calls are more likely to churn Churn vs Payment Delay: customer with 20 or more payment delays are more likely to churn Churn vs Total Spend: customer who spend more (500 above are not likely to churn) Churn vs Contract Length: customer who paid monthly are more likely to churn

```
sns.pairplot(
    train[num_cols + ['Churn']],
    hue='Churn',
    plot_kws={'alpha': 0.4},
    diag_kind='kde'
)
```

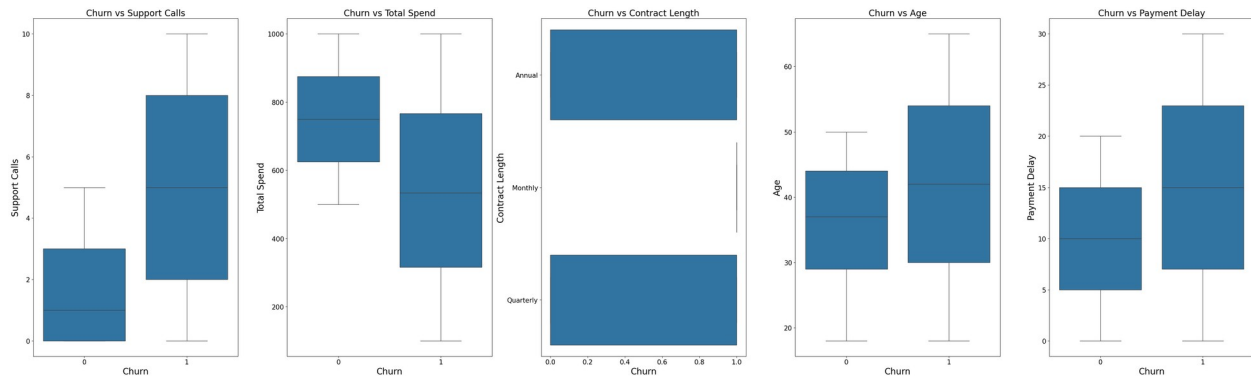
```
<seaborn.axisgrid.PairGrid at 0x7c8c4dce0860>
```



```
# checking for outliers
fig, axes = plt.subplots(1,len(features),figsize=(40,12))

for i,feature in enumerate(features):
    sns.boxplot(data=train, x='Churn', y=feature, ax=axes[i])
    axes[i].set_title(f'Churn vs {feature}',fontsize=20)
    axes[i].set_xlabel('Churn',fontsize=20)
    axes[i].set_ylabel(feature,fontsize=20)
    axes[i].tick_params(axis='both',labelsize=15)

plt.tight_layout()
```



Phase 4: Data Preprocessing

```
df = pd.concat([train,test], ignore_index=True)
df.shape
(505206, 6)
```

Handling Outliers

looking on the boxplot earlier, it shows no outlier indicator so we'll skip the process for cleaning the outliers

Encoding Categorical Columns

```
# encoding categorical variables
from sklearn.preprocessing import OrdinalEncoder

oe = OrdinalEncoder(categories=[['Monthly', 'Quarterly', 'Annual']])
df[['Contract Length']] = oe.fit_transform(df[['Contract Length']])

#checking
df.head()

{"type": "dataframe", "variable_name": "df"}

# checking for correct dtype
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 505206 entries, 0 to 505205
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                    505206 non-null  int64
1   Support Calls          505206 non-null  int64
2   Payment Delay          505206 non-null  int64
3   Contract Length        505206 non-null  float64
4   Total Spend            505206 non-null  float64
```

```

5    Churn          505206 non-null  int64
dtypes: float64(2), int64(4)
memory usage: 23.1 MB

# changing the dtype of Contract Length Column
df['Contract Length'] = df['Contract Length'].astype(int)

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 505206 entries, 0 to 505205
Data columns (total 6 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                   505206 non-null  int64
1   Support Calls         505206 non-null  int64
2   Payment Delay         505206 non-null  int64
3   Contract Length       505206 non-null  int64
4   Total Spend           505206 non-null  float64
5   Churn                 505206 non-null  int64
dtypes: float64(1), int64(5)
memory usage: 23.1 MB

```

Scaling

since we are using XGBoost, LightGBM and CatBoost model, we will not need to scale our data

Feature Engineering

```

# feature engineering we will add columns to help more of the model we
will train
# adding risk_score column by combining columns with high correlation
df['risk_score'] = df['Payment Delay'] * df['Support Calls']

# adding high caller flag to know the model if the customer is a high
caller
df['high_caller_flag'] = (df['Support Calls'] >= 4).astype(int)

# adding a low spender flag to know if the customer spends less money
df['low_spender_flag'] = (df['Total Spend'] < 500).astype(int)

df.head()

{"type": "dataframe", "variable_name": "df"}

# move churn column to the end
churn_col = df.pop('Churn')
df['Churn'] = churn_col

# checking
df.head()

```

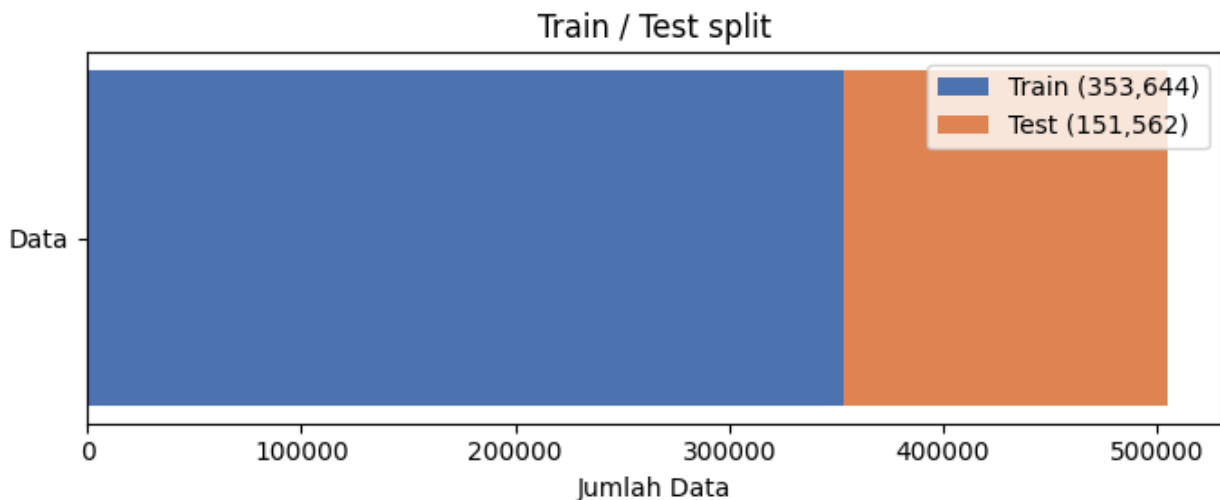
```
{"type": "dataframe", "variable_name": "df"}
```

Model Training

```
# removing the targets
X = df.drop('Churn', axis=1)
y = df['Churn']

# splitting the data
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.3, random_state=42)

fig, ax = plt.subplots(figsize=(7, 3))
ax.barh(['Data'], [len(X_train)], color='#4C72B0', label=f'Train
({len(X_train):,})')
ax.barh(['Data'], [len(X_test)], left=[len(X_train)], color='#DD8452',
label=f'Test ({len(X_test):,})')
ax.set_xlabel('Number of records')
ax.set_title('Train / Test split')
ax.legend()
plt.tight_layout()
```



```
!pip install catboost
```

Collecting catboost

Downloading catboost-1.2.10-cp312-cp312-manylinux2014_x86_64.whl.metadata (1.2 kB)

Requirement already satisfied: graphviz in /usr/local/lib/python3.12/dist-packages (from catboost) (0.21)

Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from catboost) (3.10.0)

Requirement already satisfied: numpy<3.0,>=1.16.0 in

```
/usr/local/lib/python3.12/dist-packages (from catboost) (2.0.2)
Requirement already satisfied: pandas<4.0,>=0.24 in
/usr/local/lib/python3.12/dist-packages (from catboost) (2.2.2)
Requirement already satisfied: scipy in
/usr/local/lib/python3.12/dist-packages (from catboost) (1.16.3)
Requirement already satisfied: plotly in
/usr/local/lib/python3.12/dist-packages (from catboost) (5.24.1)
Requirement already satisfied: six in /usr/local/lib/python3.12/dist-
packages (from catboost) (1.17.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from pandas<4.0,>=0.24-
>catboost) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
/usr/local/lib/python3.12/dist-packages (from pandas<4.0,>=0.24-
>catboost) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in
/usr/local/lib/python3.12/dist-packages (from pandas<4.0,>=0.24-
>catboost) (2026.2)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost)
(0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost)
(4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost)
(1.5.0)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost)
(26.2)
Requirement already satisfied: pillow>=8 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost)
(11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib->catboost)
(3.3.2)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->catboost)
(9.1.4)
Downloading catboost-1.2.10-cp312-cp312-manylinux2014_x86_64.whl (97.1
MB)
----- 97.1/97.1 MB 8.9 MB/s eta
0:00:00
```


Doing Hyperparameter tuning

```
!pip install optuna
```

```
Collecting optuna
```

```
  Downloading optuna-4.8.0-py3-none-any.whl.metadata (17 kB)
```

```
Requirement already satisfied: alembic>=1.5.0 in  
/usr/local/lib/python3.12/dist-packages (from optuna) (1.18.4)
```

```
Collecting colorlog (from optuna)
```

```
  Downloading colorlog-6.10.1-py3-none-any.whl.metadata (11 kB)
```

```
Requirement already satisfied: numpy in
```

```
/usr/local/lib/python3.12/dist-packages (from optuna) (2.0.2)
```

```
Requirement already satisfied: packaging>=20.0 in
```

```
/usr/local/lib/python3.12/dist-packages (from optuna) (26.2)
```

```
Requirement already satisfied: sqlalchemy>=1.4.2 in
```

```
/usr/local/lib/python3.12/dist-packages (from optuna) (2.0.49)
```

```
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-  
packages (from optuna) (4.67.3)
```

```
Requirement already satisfied: PyYAML in
```

```
/usr/local/lib/python3.12/dist-packages (from optuna) (6.0.3)
```

```
Requirement already satisfied: Mako in /usr/local/lib/python3.12/dist-  
packages (from alembic>=1.5.0->optuna) (1.3.12)
```

```
Requirement already satisfied: typing-extensions>=4.12 in
```

```
/usr/local/lib/python3.12/dist-packages (from alembic>=1.5.0->optuna)  
(4.15.0)
```

```
Requirement already satisfied: greenlet>=1 in
```

```
/usr/local/lib/python3.12/dist-packages (from sqlalchemy>=1.4.2-  
>optuna) (3.5.1)
```

```
Requirement already satisfied: MarkupSafe>=0.9.2 in
```

```
/usr/local/lib/python3.12/dist-packages (from Mako->alembic>=1.5.0-  
>optuna) (3.0.3)
```

```
Downloading optuna-4.8.0-py3-none-any.whl (419 kB)
```

```
419.5/419.5 kB 10.4 MB/s eta
```

```
0:00:00
```

```
import optuna
```

```
from sklearn.model_selection import cross_val_score, StratifiedKFold
```

```
from xgboost import XGBClassifier
```

```
from lightgbm import LGBMClassifier
```

```
from catboost import CatBoostClassifier
```

```
def objective(trial):
```

```
    params = {
```

```
        'n_estimators'      : trial.suggest_int('n_estimators', 50,  
300),
```

```
        'max_depth'         : trial.suggest_int('max_depth', 2, 6),
```

```
        'learning_rate'     : trial.suggest_float('learning_rate', 0.01,  
0.3),
```

```
        'reg_alpha'         : trial.suggest_float('reg_alpha', 0, 1),
```

```
        'reg_lambda'        : trial.suggest_float('reg_lambda', 0, 1),
```



```

        'min_child_weight': trial.suggest_int('min_child_weight', 1,
10),
        'random_state'      : 42,
        'eval_metric'       : 'logloss'
    }
    model = XGBClassifier(**params)
    skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
    scores = cross_val_score(model, X_test, y_test, cv=skf,
scoring='accuracy')
    return scores.mean()

```

```

study_XGB = optuna.create_study(direction='maximize')
study_XGB.optimize(objective, n_trials=50)
print("Best params:", study_XGB.best_params)

```

```

[I 2026-05-30 03:04:50,151] A new study created in memory with name:
no-name-16c4e235-1f56-4c24-8280-f06e5dfd4da8
[I 2026-05-30 03:05:14,568] Trial 0 finished with value:
0.9213786991474896 and parameters: {'n_estimators': 229, 'max_depth':
3, 'learning_rate': 0.0877731637298579, 'reg_alpha':
0.443247712205745, 'reg_lambda': 0.7525811152501012,
'min_child_weight': 2}. Best is trial 0 with value:
0.9213786991474896.
[I 2026-05-30 03:05:17,003] Trial 1 finished with value:
0.9211015848746966 and parameters: {'n_estimators': 70, 'max_depth':
3, 'learning_rate': 0.164802808206593, 'reg_alpha':
0.9533370198763513, 'reg_lambda': 0.19167513564933059,
'min_child_weight': 10}. Best is trial 0 with value:
0.9213786991474896.
[I 2026-05-30 03:05:21,342] Trial 2 finished with value:
0.9244863398769991 and parameters: {'n_estimators': 139, 'max_depth':
4, 'learning_rate': 0.18137755150010565, 'reg_alpha':
0.886673861130115, 'reg_lambda': 0.8003726197842543,
'min_child_weight': 5}. Best is trial 2 with value:
0.9244863398769991.
[I 2026-05-30 03:05:28,780] Trial 3 finished with value:
0.9211807581734746 and parameters: {'n_estimators': 168, 'max_depth':
3, 'learning_rate': 0.09274814526306235, 'reg_alpha':
0.9877156895818777, 'reg_lambda': 0.717936722437484,
'min_child_weight': 5}. Best is trial 2 with value:
0.9244863398769991.
[I 2026-05-30 03:05:30,889] Trial 4 finished with value:
0.9175518937167535 and parameters: {'n_estimators': 75, 'max_depth':
2, 'learning_rate': 0.213103641067374, 'reg_alpha':
0.39079443858598295, 'reg_lambda': 0.49090186715673245,
'min_child_weight': 9}. Best is trial 2 with value:
0.9244863398769991.
[I 2026-05-30 03:05:33,347] Trial 5 finished with value:
0.9011757931719642 and parameters: {'n_estimators': 91, 'max_depth':
2, 'learning_rate': 0.06681523717684561, 'reg_alpha':

```

0.44011884923223477, 'reg_lambda': 0.6077249874457314,
'min_child_weight': 1}. Best is trial 2 with value:
0.9244863398769991.
[I 2026-05-30 03:05:36,264] Trial 6 finished with value:
0.9232459242405685 and parameters: {'n_estimators': 80, 'max_depth':
4, 'learning_rate': 0.1339127050967401, 'reg_alpha':
0.7918309953411709, 'reg_lambda': 0.805380260737776,
'min_child_weight': 2}. Best is trial 2 with value:
0.9244863398769991.
[I 2026-05-30 03:05:49,334] Trial 7 finished with value:
0.9260104652212335 and parameters: {'n_estimators': 300, 'max_depth':
6, 'learning_rate': 0.21851762848341355, 'reg_alpha':
0.5266578998850281, 'reg_lambda': 0.13654308802144488,
'min_child_weight': 10}. Best is trial 7 with value:
0.9260104652212335.
[I 2026-05-30 03:05:54,622] Trial 8 finished with value:
0.9261292281078639 and parameters: {'n_estimators': 67, 'max_depth':
5, 'learning_rate': 0.29996910591410714, 'reg_alpha':
0.7271377329513993, 'reg_lambda': 0.022143009337665442,
'min_child_weight': 1}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:05:58,749] Trial 9 finished with value:
0.9183238377875341 and parameters: {'n_estimators': 138, 'max_depth':
3, 'learning_rate': 0.04359250843688861, 'reg_alpha':
0.82320428836871, 'reg_lambda': 0.37135667964977437,
'min_child_weight': 10}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:06:09,668] Trial 10 finished with value:
0.9257597422656973 and parameters: {'n_estimators': 232, 'max_depth':
6, 'learning_rate': 0.28656608610056267, 'reg_alpha':
0.08388882016461052, 'reg_lambda': 0.00950728041519643,
'min_child_weight': 7}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:06:22,635] Trial 11 finished with value:
0.9253044820304792 and parameters: {'n_estimators': 280, 'max_depth':
6, 'learning_rate': 0.29621744273190936, 'reg_alpha':
0.6021554047312688, 'reg_lambda': 0.020533170206368967,
'min_child_weight': 7}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:06:36,758] Trial 12 finished with value:
0.9258851040699614 and parameters: {'n_estimators': 294, 'max_depth':
5, 'learning_rate': 0.23921151940597868, 'reg_alpha':
0.6406833492396551, 'reg_lambda': 0.2318890772959324,
'min_child_weight': 4}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:06:57,419] Trial 13 finished with value:
0.9260566502440966 and parameters: {'n_estimators': 236, 'max_depth':
5, 'learning_rate': 0.24552857560520924, 'reg_alpha':
0.2036448134717564, 'reg_lambda': 0.9996717832399455,

'min_child_weight': 8}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:07:11,929] Trial 14 finished with value:
0.9258982992932616 and parameters: {'n_estimators': 215, 'max_depth':
5, 'learning_rate': 0.25871834863537324, 'reg_alpha':
0.21632159302345072, 'reg_lambda': 0.899631712507895,
'min_child_weight': 7}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:07:26,640] Trial 15 finished with value:
0.9260038660859358 and parameters: {'n_estimators': 201, 'max_depth':
5, 'learning_rate': 0.25677955445768397, 'reg_alpha':
0.2768072185689231, 'reg_lambda': 0.985593007786543,
'min_child_weight': 8}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:07:48,469] Trial 16 finished with value:
0.9257201548544843 and parameters: {'n_estimators': 258, 'max_depth':
5, 'learning_rate': 0.2823026072807866, 'reg_alpha':
0.006995624519399057, 'reg_lambda': 0.3596118584230372,
'min_child_weight': 3}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:07:56,102] Trial 17 finished with value:
0.9246182931983231 and parameters: {'n_estimators': 112, 'max_depth':
4, 'learning_rate': 0.19781546602374384, 'reg_alpha':
0.26298725464180334, 'reg_lambda': 0.6140469896086252,
'min_child_weight': 6}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:08:02,949] Trial 18 finished with value:
0.9258191225118606 and parameters: {'n_estimators': 51, 'max_depth':
5, 'learning_rate': 0.1383546381644654, 'reg_alpha':
0.677098734431034, 'reg_lambda': 0.3949013010700249,
'min_child_weight': 8}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:08:09,969] Trial 19 finished with value:
0.9259180925635402 and parameters: {'n_estimators': 189, 'max_depth':
6, 'learning_rate': 0.24643283929244947, 'reg_alpha':
0.737838993586565, 'reg_lambda': 0.5448799918472126,
'min_child_weight': 1}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:08:18,202] Trial 20 finished with value:
0.9260104645682418 and parameters: {'n_estimators': 165, 'max_depth':
5, 'learning_rate': 0.27317511772037756, 'reg_alpha':
0.19210330293259414, 'reg_lambda': 0.989189311389402,
'min_child_weight': 4}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:08:30,768] Trial 21 finished with value:
0.9260764446026949 and parameters: {'n_estimators': 259, 'max_depth':
6, 'learning_rate': 0.22044567299507775, 'reg_alpha':
0.5617514451140652, 'reg_lambda': 0.14318847122225803,
'min_child_weight': 9}. Best is trial 8 with value:

0.9261292281078639.
[I 2026-05-30 03:08:43,381] Trial 22 finished with value:
0.9260830417790171 and parameters: {'n_estimators': 261, 'max_depth':
6, 'learning_rate': 0.22739698922182328, 'reg_alpha':
0.5587812556086944, 'reg_lambda': 0.1165393864565978,
'min_child_weight': 9}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:08:52,766] Trial 23 finished with value:
0.9260038680449112 and parameters: {'n_estimators': 261, 'max_depth':
6, 'learning_rate': 0.22440952579644508, 'reg_alpha':
0.5530165808904269, 'reg_lambda': 0.1024830562874327,
'min_child_weight': 9}. Best is trial 8 with value:
0.9261292281078639.
[I 2026-05-30 03:09:04,567] Trial 24 finished with value:
0.9261886096600108 and parameters: {'n_estimators': 263, 'max_depth':
6, 'learning_rate': 0.1685993875041504, 'reg_alpha':
0.6924161239726718, 'reg_lambda': 0.2725210515345693,
'min_child_weight': 9}. Best is trial 24 with value:
0.9261886096600108.
[I 2026-05-30 03:09:17,204] Trial 25 finished with value:
0.9260962383083011 and parameters: {'n_estimators': 275, 'max_depth':
6, 'learning_rate': 0.1809630151138861, 'reg_alpha':
0.7003092665214555, 'reg_lambda': 0.2610075312493069,
'min_child_weight': 6}. Best is trial 24 with value:
0.9261886096600108.
[I 2026-05-30 03:09:25,420] Trial 26 finished with value:
0.9262545883884803 and parameters: {'n_estimators': 141, 'max_depth':
6, 'learning_rate': 0.12116458796562897, 'reg_alpha':
0.74394749867884, 'reg_lambda': 0.21139815168319745,
'min_child_weight': 4}. Best is trial 26 with value:
0.9262545883884803.
[I 2026-05-30 03:09:30,343] Trial 27 finished with value:
0.9168657020549661 and parameters: {'n_estimators': 141, 'max_depth':
4, 'learning_rate': 0.01913910374098371, 'reg_alpha':
0.809155604470066, 'reg_lambda': 0.2905562601913586,
'min_child_weight': 3}. Best is trial 26 with value:
0.9262545883884803.
[I 2026-05-30 03:09:35,410] Trial 28 finished with value:
0.9260434543678043 and parameters: {'n_estimators': 111, 'max_depth':
5, 'learning_rate': 0.11964105777403881, 'reg_alpha':
0.749982633501596, 'reg_lambda': 0.3095166343537745,
'min_child_weight': 2}. Best is trial 26 with value:
0.9262545883884803.
[I 2026-05-30 03:09:42,027] Trial 29 finished with value:
0.9261490211604786 and parameters: {'n_estimators': 114, 'max_depth':
6, 'learning_rate': 0.09725916799777898, 'reg_alpha':
0.9021027479741321, 'reg_lambda': 0.050983688895184856,
'min_child_weight': 4}. Best is trial 26 with value:
0.9262545883884803.

[I 2026-05-30 03:09:46,532] Trial 30 finished with value: 0.9260566493734409 and parameters: {'n_estimators': 108, 'max_depth': 6, 'learning_rate': 0.09938152799279051, 'reg_alpha': 0.8704485078866336, 'reg_lambda': 0.42419186802612047, 'min_child_weight': 4}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:09:55,701] Trial 31 finished with value: 0.926188609442347 and parameters: {'n_estimators': 157, 'max_depth': 6, 'learning_rate': 0.059703968864006106, 'reg_alpha': 0.8953099664399347, 'reg_lambda': 0.059058137826513105, 'min_child_weight': 3}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:10:02,224] Trial 32 finished with value: 0.9261226289725661 and parameters: {'n_estimators': 154, 'max_depth': 6, 'learning_rate': 0.06625131843757069, 'reg_alpha': 0.9259564915005151, 'reg_lambda': 0.20696680448307692, 'min_child_weight': 3}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:10:10,638] Trial 33 finished with value: 0.9261952081423169 and parameters: {'n_estimators': 129, 'max_depth': 6, 'learning_rate': 0.11390838123917929, 'reg_alpha': 0.879505682287325, 'reg_lambda': 0.07416017198002506, 'min_child_weight': 5}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:10:20,178] Trial 34 finished with value: 0.92624139316518 and parameters: {'n_estimators': 184, 'max_depth': 6, 'learning_rate': 0.1622816154450346, 'reg_alpha': 0.9897572850214378, 'reg_lambda': 0.1754935463261253, 'min_child_weight': 5}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:10:27,162] Trial 35 finished with value: 0.9262281981595434 and parameters: {'n_estimators': 185, 'max_depth': 6, 'learning_rate': 0.15239227063665406, 'reg_alpha': 0.8413557067131228, 'reg_lambda': 0.17606828290882556, 'min_child_weight': 5}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:10:37,203] Trial 36 finished with value: 0.926241393382844 and parameters: {'n_estimators': 196, 'max_depth': 6, 'learning_rate': 0.14204544354687323, 'reg_alpha': 0.9700911195530579, 'reg_lambda': 0.18254620918948006, 'min_child_weight': 5}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:10:42,130] Trial 37 finished with value: 0.9216030301327776 and parameters: {'n_estimators': 182, 'max_depth': 3, 'learning_rate': 0.15081850260220397, 'reg_alpha': 0.9761849071621944, 'reg_lambda': 0.16762565636051158, 'min_child_weight': 5}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:10:51,351] Trial 38 finished with value: 0.9261358248488584 and parameters: {'n_estimators': 200, 'max_depth':

5, 'learning_rate': 0.15554216296173995, 'reg_alpha': 0.9863275443125026, 'reg_lambda': 0.19442085869571404, 'min_child_weight': 6}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:10:58,539] Trial 39 finished with value: 0.9261754142190466 and parameters: {'n_estimators': 176, 'max_depth': 6, 'learning_rate': 0.19835230441594193, 'reg_alpha': 0.8331732554277856, 'reg_lambda': 0.46809529901388547, 'min_child_weight': 5}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:11:05,871] Trial 40 finished with value: 0.9185415746437948 and parameters: {'n_estimators': 220, 'max_depth': 2, 'learning_rate': 0.13361471985069295, 'reg_alpha': 0.7755125588255867, 'reg_lambda': 0.3436606441733109, 'min_child_weight': 6}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:11:11,234] Trial 41 finished with value: 0.926208404671601 and parameters: {'n_estimators': 131, 'max_depth': 6, 'learning_rate': 0.11567709538905482, 'reg_alpha': 0.9418023361198653, 'reg_lambda': 0.10351800686244814, 'min_child_weight': 5}. Best is trial 26 with value: 0.9262545883884803.

[I 2026-05-30 03:11:21,358] Trial 42 finished with value: 0.9262743829647425 and parameters: {'n_estimators': 197, 'max_depth': 6, 'learning_rate': 0.10976625514053143, 'reg_alpha': 0.9456669164402142, 'reg_lambda': 0.21994452195209485, 'min_child_weight': 4}. Best is trial 42 with value: 0.9262743829647425.

[I 2026-05-30 03:11:31,612] Trial 43 finished with value: 0.9262809812293845 and parameters: {'n_estimators': 196, 'max_depth': 6, 'learning_rate': 0.08001907994392687, 'reg_alpha': 0.8460001270228473, 'reg_lambda': 0.22400112377335663, 'min_child_weight': 4}. Best is trial 43 with value: 0.9262809812293845.

[I 2026-05-30 03:11:39,430] Trial 44 finished with value: 0.9262281972888877 and parameters: {'n_estimators': 195, 'max_depth': 6, 'learning_rate': 0.07555213967426706, 'reg_alpha': 0.9571859413961333, 'reg_lambda': 0.23977555545657706, 'min_child_weight': 4}. Best is trial 43 with value: 0.9262809812293845.

[I 2026-05-30 03:11:49,624] Trial 45 finished with value: 0.9258587105760652 and parameters: {'n_estimators': 209, 'max_depth': 5, 'learning_rate': 0.0466471834898231, 'reg_alpha': 0.9877267770299887, 'reg_lambda': 0.32577160525123816, 'min_child_weight': 4}. Best is trial 43 with value: 0.9262809812293845.

[I 2026-05-30 03:11:58,840] Trial 46 finished with value: 0.9262150007596036 and parameters: {'n_estimators': 166, 'max_depth': 6, 'learning_rate': 0.0836785009007635, 'reg_alpha':

```

0.4504601511146715, 'reg_lambda': 0.21718140186881002,
'min_child_weight': 3}. Best is trial 43 with value:
0.9262809812293845.
[I 2026-05-30 03:12:05,108] Trial 47 finished with value:
0.9244269613721473 and parameters: {'n_estimators': 211, 'max_depth':
4, 'learning_rate': 0.10685850547456711, 'reg_alpha':
0.9144950416052817, 'reg_lambda': 0.1561919909640342,
'min_child_weight': 4}. Best is trial 43 with value:
0.9262809812293845.
[I 2026-05-30 03:12:16,609] Trial 48 finished with value:
0.9262743834000705 and parameters: {'n_estimators': 242, 'max_depth':
6, 'learning_rate': 0.12877667936283707, 'reg_alpha':
0.8410262609359906, 'reg_lambda': 0.43135485639183196,
'min_child_weight': 5}. Best is trial 43 with value:
0.9262809812293845.
[I 2026-05-30 03:12:27,120] Trial 49 finished with value:
0.9260434541501403 and parameters: {'n_estimators': 242, 'max_depth':
5, 'learning_rate': 0.12863546364244366, 'reg_alpha':
0.785842347516312, 'reg_lambda': 0.6419621699372982,
'min_child_weight': 2}. Best is trial 43 with value:
0.9262809812293845.

```

```

Best params: {'n_estimators': 196, 'max_depth': 6, 'learning_rate':
0.08001907994392687, 'reg_alpha': 0.8460001270228473, 'reg_lambda':
0.22400112377335663, 'min_child_weight': 4}

```

```

def objective_lgbm(trial):
    params = {
        'n_estimators'      : trial.suggest_int('n_estimators', 50,
500),
        'max_depth'         : trial.suggest_int('max_depth', 2, 8),
        'learning_rate'     : trial.suggest_float('learning_rate',
0.01, 0.3, log=True),
        'subsample'         : trial.suggest_float('subsample', 0.5,
1.0),
        'colsample_bytree'  : trial.suggest_float('colsample_bytree',
0.5, 1.0),
        'reg_alpha'         : trial.suggest_float('reg_alpha', 0.0,
1.0),
        'reg_lambda'        : trial.suggest_float('reg_lambda', 0.0,
1.0),
        'min_child_samples' : trial.suggest_int('min_child_samples',
5, 50),
        'num_leaves'        : trial.suggest_int('num_leaves', 20,
150),
        'random_state'      : 42,
        'verbose'           : -1
    }
    model = LGBMClassifier(**params)
    skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

```

```
scores = cross_val_score(model, X_train, y_train, cv=skf,  
scoring='accuracy')  
return scores.mean()
```

```
study_LGBM = optuna.create_study(direction='maximize')  
study_LGBM.optimize(objective_lgbm, n_trials=50)  
print("Best params:", study_LGBM.best_params)
```

```
[I 2026-05-30 03:13:03,798] A new study created in memory with name:  
no-name-4d7f22ff-5bab-45c3-bb9a-d617e42a32f0  
[I 2026-05-30 03:13:42,417] Trial 0 finished with value:  
0.9283856076516331 and parameters: {'n_estimators': 235, 'max_depth':  
6, 'learning_rate': 0.033512098153126134, 'subsample':  
0.6946795925396857, 'colsample_bytree': 0.9346800427345581,  
'reg_alpha': 0.8807571566234942, 'reg_lambda': 0.9809409048149828,  
'min_child_samples': 32, 'num_leaves': 66}. Best is trial 0 with  
value: 0.9283856076516331.  
[I 2026-05-30 03:14:14,731] Trial 1 finished with value:  
0.9121574262859344 and parameters: {'n_estimators': 212, 'max_depth':  
6, 'learning_rate': 0.013095715544592943, 'subsample':  
0.6201242650724758, 'colsample_bytree': 0.511090343554748,  
'reg_alpha': 0.5922507845388107, 'reg_lambda': 0.32450975595793885,  
'min_child_samples': 17, 'num_leaves': 144}. Best is trial 0 with  
value: 0.9283856076516331.  
[I 2026-05-30 03:14:32,676] Trial 2 finished with value:  
0.9210448931370662 and parameters: {'n_estimators': 136, 'max_depth':  
5, 'learning_rate': 0.011565083022506972, 'subsample':  
0.7147702235574589, 'colsample_bytree': 0.6425698885326716,  
'reg_alpha': 0.9728390850253836, 'reg_lambda': 0.06756812094066988,  
'min_child_samples': 47, 'num_leaves': 129}. Best is trial 0 with  
value: 0.9283856076516331.  
[I 2026-05-30 03:14:44,704] Trial 3 finished with value:  
0.9285552698777112 and parameters: {'n_estimators': 75, 'max_depth':  
6, 'learning_rate': 0.23754101164148195, 'subsample':  
0.8871353628377057, 'colsample_bytree': 0.6963488678154348,  
'reg_alpha': 0.7805895099628695, 'reg_lambda': 0.31924324569090556,  
'min_child_samples': 33, 'num_leaves': 102}. Best is trial 3 with  
value: 0.9285552698777112.  
[I 2026-05-30 03:15:18,329] Trial 4 finished with value:  
0.9285043714177827 and parameters: {'n_estimators': 284, 'max_depth':  
5, 'learning_rate': 0.2119827566744531, 'subsample':  
0.9076581145020328, 'colsample_bytree': 0.9035189991626779,  
'reg_alpha': 0.8296623485858585, 'reg_lambda': 0.652296615852781,  
'min_child_samples': 38, 'num_leaves': 115}. Best is trial 3 with  
value: 0.9285552698777112.  
[I 2026-05-30 03:15:41,042] Trial 5 finished with value:  
0.9207762612857724 and parameters: {'n_estimators': 258, 'max_depth':  
2, 'learning_rate': 0.1013227456007379, 'subsample':  
0.5386248466828917, 'colsample_bytree': 0.7531666100649503,  
'reg_alpha': 0.1269361090536818, 'reg_lambda': 0.47923438258710505,
```


'min_child_samples': 40, 'num_leaves': 97}. Best is trial 3 with value: 0.9285552698777112.

[I 2026-05-30 03:16:05,998] Trial 6 finished with value: 0.9213870449638231 and parameters: {'n_estimators': 208, 'max_depth': 4, 'learning_rate': 0.017357768875653743, 'subsample': 0.9334901333605012, 'colsample_bytree': 0.9776087513999767, 'reg_alpha': 0.4019151139173598, 'reg_lambda': 0.06507641618230686, 'min_child_samples': 22, 'num_leaves': 75}. Best is trial 3 with value: 0.9285552698777112.

[I 2026-05-30 03:16:26,506] Trial 7 finished with value: 0.92848457691703 and parameters: {'n_estimators': 130, 'max_depth': 7, 'learning_rate': 0.24296436024658394, 'subsample': 0.8201371602653229, 'colsample_bytree': 0.5313331707983819, 'reg_alpha': 0.4739252997080239, 'reg_lambda': 0.6082778609326611, 'min_child_samples': 27, 'num_leaves': 131}. Best is trial 3 with value: 0.9285552698777112.

[I 2026-05-30 03:17:13,483] Trial 8 finished with value: 0.923931977435459 and parameters: {'n_estimators': 307, 'max_depth': 6, 'learning_rate': 0.01005704416031459, 'subsample': 0.9151113477809082, 'colsample_bytree': 0.7330426181262967, 'reg_alpha': 0.20798618121463242, 'reg_lambda': 0.8322754232949147, 'min_child_samples': 11, 'num_leaves': 121}. Best is trial 3 with value: 0.9285552698777112.

[I 2026-05-30 03:17:46,980] Trial 9 finished with value: 0.9055066816083179 and parameters: {'n_estimators': 360, 'max_depth': 2, 'learning_rate': 0.021282817351708665, 'subsample': 0.5386136389699477, 'colsample_bytree': 0.8612952101805966, 'reg_alpha': 0.3837534321964735, 'reg_lambda': 0.23647100433193358, 'min_child_samples': 22, 'num_leaves': 64}. Best is trial 3 with value: 0.9285552698777112.

[I 2026-05-30 03:18:31,728] Trial 10 finished with value: 0.9285835465022659 and parameters: {'n_estimators': 477, 'max_depth': 8, 'learning_rate': 0.08991073812480135, 'subsample': 0.9927206279809131, 'colsample_bytree': 0.6438915245529735, 'reg_alpha': 0.7117212566922533, 'reg_lambda': 0.3452907832183638, 'min_child_samples': 48, 'num_leaves': 23}. Best is trial 10 with value: 0.9285835465022659.

[I 2026-05-30 03:19:16,699] Trial 11 finished with value: 0.9286174793552758 and parameters: {'n_estimators': 494, 'max_depth': 8, 'learning_rate': 0.10082953451824217, 'subsample': 0.8432191084710191, 'colsample_bytree': 0.6531001679405505, 'reg_alpha': 0.7035568103689249, 'reg_lambda': 0.3643773918262202, 'min_child_samples': 50, 'num_leaves': 24}. Best is trial 11 with value: 0.9286174793552758.

[I 2026-05-30 03:20:01,758] Trial 12 finished with value: 0.9285665806554686 and parameters: {'n_estimators': 493, 'max_depth': 8, 'learning_rate': 0.08287009217354985, 'subsample': 0.8156766245167999, 'colsample_bytree': 0.6092231310356165, 'reg_alpha': 0.635067471400315, 'reg_lambda': 0.4832475643478547,

'min_child_samples': 50, 'num_leaves': 20}. Best is trial 11 with value: 0.9286174793552758.

[I 2026-05-30 03:20:47,279] Trial 13 finished with value: 0.9286740333640022 and parameters: {'n_estimators': 487, 'max_depth': 8, 'learning_rate': 0.11356714867241353, 'subsample': 0.9913314605982633, 'colsample_bytree': 0.7851201202096073, 'reg_alpha': 0.7011203982888321, 'reg_lambda': 0.23305037893427305, 'min_child_samples': 44, 'num_leaves': 25}. Best is trial 13 with value: 0.9286740333640022.

[I 2026-05-30 03:21:26,346] Trial 14 finished with value: 0.9286768610184616 and parameters: {'n_estimators': 402, 'max_depth': 8, 'learning_rate': 0.1359553501137728, 'subsample': 0.9971928281461986, 'colsample_bytree': 0.8277637868965072, 'reg_alpha': 0.5865825186167285, 'reg_lambda': 0.16807657975221663, 'min_child_samples': 41, 'num_leaves': 42}. Best is trial 14 with value: 0.9286768610184616.

[I 2026-05-30 03:22:19,741] Trial 15 finished with value: 0.9286174791553765 and parameters: {'n_estimators': 404, 'max_depth': 7, 'learning_rate': 0.15075648611662537, 'subsample': 0.9733599862386142, 'colsample_bytree': 0.8109960886311948, 'reg_alpha': 0.5492611716203845, 'reg_lambda': 0.1933617943091852, 'min_child_samples': 41, 'num_leaves': 47}. Best is trial 14 with value: 0.9286768610184616.

[I 2026-05-30 03:23:23,625] Trial 16 finished with value: 0.9286881719161585 and parameters: {'n_estimators': 415, 'max_depth': 7, 'learning_rate': 0.05355479068659683, 'subsample': 0.997708169255516, 'colsample_bytree': 0.8109182528142362, 'reg_alpha': 0.32671630109724925, 'reg_lambda': 0.012602167784174056, 'min_child_samples': 43, 'num_leaves': 44}. Best is trial 16 with value: 0.9286881719161585.

[I 2026-05-30 03:24:10,779] Trial 17 finished with value: 0.9286909995306383 and parameters: {'n_estimators': 405, 'max_depth': 7, 'learning_rate': 0.05305430602662088, 'subsample': 0.7719509698756228, 'colsample_bytree': 0.8297457650301542, 'reg_alpha': 0.22542075719321558, 'reg_lambda': 0.007384910035073883, 'min_child_samples': 35, 'num_leaves': 46}. Best is trial 17 with value: 0.9286909995306383.

[I 2026-05-30 03:24:50,641] Trial 18 finished with value: 0.9255974933382743 and parameters: {'n_estimators': 358, 'max_depth': 4, 'learning_rate': 0.05035702139748505, 'subsample': 0.7656706895706373, 'colsample_bytree': 0.871218685296126, 'reg_alpha': 0.2471070200625472, 'reg_lambda': 0.008924521218586124, 'min_child_samples': 34, 'num_leaves': 46}. Best is trial 17 with value: 0.9286909995306383.

[I 2026-05-30 03:25:52,722] Trial 19 finished with value: 0.9286825164073405 and parameters: {'n_estimators': 424, 'max_depth': 7, 'learning_rate': 0.052808034443092955, 'subsample': 0.6333512984315812, 'colsample_bytree': 0.9902433805950771, 'reg_alpha': 0.015654737279226727, 'reg_lambda': 0.11917360554487984,

'min_child_samples': 27, 'num_leaves': 54}. Best is trial 17 with value: 0.9286909995306383.

[I 2026-05-30 03:26:31,463] Trial 20 finished with value: 0.9243787539164915 and parameters: {'n_estimators': 335, 'max_depth': 4, 'learning_rate': 0.04076940804233081, 'subsample': 0.7754458823828368, 'colsample_bytree': 0.7420148559029274, 'reg_alpha': 0.25086831135817445, 'reg_lambda': 0.03733394828252441, 'min_child_samples': 36, 'num_leaves': 36}. Best is trial 17 with value: 0.9286909995306383.

[I 2026-05-30 03:27:22,816] Trial 21 finished with value: 0.9286768609784819 and parameters: {'n_estimators': 431, 'max_depth': 7, 'learning_rate': 0.0583355736464736, 'subsample': 0.6190562765901385, 'colsample_bytree': 0.9894782625100174, 'reg_alpha': 0.018583663619197444, 'reg_lambda': 0.14745241702494857, 'min_child_samples': 29, 'num_leaves': 58}. Best is trial 17 with value: 0.9286909995306383.

[I 2026-05-30 03:28:28,402] Trial 22 finished with value: 0.9286881717162594 and parameters: {'n_estimators': 430, 'max_depth': 7, 'learning_rate': 0.025707106922632016, 'subsample': 0.6142558687370877, 'colsample_bytree': 0.9301642225081084, 'reg_alpha': 0.02583578545009569, 'reg_lambda': 0.11668462445240788, 'min_child_samples': 26, 'num_leaves': 80}. Best is trial 17 with value: 0.9286909995306383.

[I 2026-05-30 03:29:32,719] Trial 23 finished with value: 0.9286966549195169 and parameters: {'n_estimators': 451, 'max_depth': 7, 'learning_rate': 0.028244587417493933, 'subsample': 0.6799526196582728, 'colsample_bytree': 0.9331316546881256, 'reg_alpha': 0.12852261041111232, 'reg_lambda': 0.000595347428729711, 'min_child_samples': 6, 'num_leaves': 81}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:30:26,132] Trial 24 finished with value: 0.9285354755368775 and parameters: {'n_estimators': 381, 'max_depth': 6, 'learning_rate': 0.032926618206641285, 'subsample': 0.6796394223528621, 'colsample_bytree': 0.863260406385692, 'reg_alpha': 0.32078214086971946, 'reg_lambda': 0.018415019606560792, 'min_child_samples': 8, 'num_leaves': 96}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:31:22,787] Trial 25 finished with value: 0.9286061686174983 and parameters: {'n_estimators': 458, 'max_depth': 7, 'learning_rate': 0.07007360245204151, 'subsample': 0.6782739002861766, 'colsample_bytree': 0.8170873570465073, 'reg_alpha': 0.12603977008929315, 'reg_lambda': 0.005617083969755213, 'min_child_samples': 6, 'num_leaves': 73}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:32:03,728] Trial 26 finished with value: 0.9277408930396049 and parameters: {'n_estimators': 322, 'max_depth': 5, 'learning_rate': 0.03908611802126765, 'subsample': 0.7318535794988644, 'colsample_bytree': 0.925075289528592, 'reg_alpha': 0.1444754578560393, 'reg_lambda': 0.2569440468482449,

'min_child_samples': 44, 'num_leaves': 36}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:32:48,908] Trial 27 finished with value: 0.9234540954360371 and parameters: {'n_estimators': 449, 'max_depth': 3, 'learning_rate': 0.02473493901859046, 'subsample': 0.5024626426624078, 'colsample_bytree': 0.7812248563250663, 'reg_alpha': 0.3303338782979182, 'reg_lambda': 0.10534246516774741, 'min_child_samples': 13, 'num_leaves': 88}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:33:40,184] Trial 28 finished with value: 0.928560925466489 and parameters: {'n_estimators': 383, 'max_depth': 6, 'learning_rate': 0.0654691169561919, 'subsample': 0.8644598029077453, 'colsample_bytree': 0.8857986668811199, 'reg_alpha': 0.48147334881297854, 'reg_lambda': 0.6004139857117796, 'min_child_samples': 20, 'num_leaves': 54}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:34:40,135] Trial 29 finished with value: 0.928688171836199 and parameters: {'n_estimators': 454, 'max_depth': 7, 'learning_rate': 0.032090728745062747, 'subsample': 0.7909864187121664, 'colsample_bytree': 0.9603344568050766, 'reg_alpha': 0.18867259820770338, 'reg_lambda': 0.8775738630689129, 'min_child_samples': 31, 'num_leaves': 65}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:35:21,917] Trial 30 finished with value: 0.9285835468620846 and parameters: {'n_estimators': 358, 'max_depth': 6, 'learning_rate': 0.04520064281474119, 'subsample': 0.664416682492745, 'colsample_bytree': 0.9456357746326416, 'reg_alpha': 0.30286893553496436, 'reg_lambda': 0.178512253922749, 'min_child_samples': 36, 'num_leaves': 33}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:36:23,572] Trial 31 finished with value: 0.9286853440617998 and parameters: {'n_estimators': 460, 'max_depth': 7, 'learning_rate': 0.030918473883444122, 'subsample': 0.7943791088211265, 'colsample_bytree': 0.9669307080055294, 'reg_alpha': 0.17065646742743576, 'reg_lambda': 0.8808103753989187, 'min_child_samples': 30, 'num_leaves': 66}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:37:28,431] Trial 32 finished with value: 0.9286938272650573 and parameters: {'n_estimators': 407, 'max_depth': 7, 'learning_rate': 0.019425321426674035, 'subsample': 0.7397036445995542, 'colsample_bytree': 0.8377866452809285, 'reg_alpha': 0.09597547257848194, 'reg_lambda': 0.989066149864641, 'min_child_samples': 16, 'num_leaves': 86}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:38:26,978] Trial 33 finished with value: 0.9284902325457878 and parameters: {'n_estimators': 403, 'max_depth': 6, 'learning_rate': 0.019899422375465933, 'subsample': 0.7395315416471463, 'colsample_bytree': 0.8293236039720898, 'reg_alpha': 0.07880445553779777, 'reg_lambda': 0.7205531040163716,

'min_child_samples': 16, 'num_leaves': 88}. Best is trial 23 with value: 0.9286966549195169.

[I 2026-05-30 03:39:38,916] Trial 34 finished with value: 0.9287107933917136 and parameters: {'n_estimators': 422, 'max_depth': 8, 'learning_rate': 0.016705329947735097, 'subsample': 0.580850086592553, 'colsample_bytree': 0.8467948701218844, 'reg_alpha': 0.08470486159663973, 'reg_lambda': 0.9632119962555444, 'min_child_samples': 5, 'num_leaves': 110}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:40:44,572] Trial 35 finished with value: 0.928685343781941 and parameters: {'n_estimators': 380, 'max_depth': 8, 'learning_rate': 0.0165871191158533, 'subsample': 0.7053377225729783, 'colsample_bytree': 0.9042674303547191, 'reg_alpha': 0.06492308614797307, 'reg_lambda': 0.9644596676471299, 'min_child_samples': 9, 'num_leaves': 106}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:41:38,844] Trial 36 finished with value: 0.9286570673173056 and parameters: {'n_estimators': 295, 'max_depth': 8, 'learning_rate': 0.014433710834248367, 'subsample': 0.6395043931658391, 'colsample_bytree': 0.9050412957425815, 'reg_alpha': 0.10247311022585473, 'reg_lambda': 0.9756858484988348, 'min_child_samples': 5, 'num_leaves': 138}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:42:05,412] Trial 37 finished with value: 0.9240450850131333 and parameters: {'n_estimators': 189, 'max_depth': 5, 'learning_rate': 0.013749422764481899, 'subsample': 0.7280491761834664, 'colsample_bytree': 0.8547902604343302, 'reg_alpha': 0.21268198125473264, 'reg_lambda': 0.753285778580745, 'min_child_samples': 15, 'num_leaves': 111}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:42:51,197] Trial 38 finished with value: 0.9286598950117451 and parameters: {'n_estimators': 239, 'max_depth': 8, 'learning_rate': 0.025297178223387073, 'subsample': 0.5798015796755004, 'colsample_bytree': 0.7756493507241046, 'reg_alpha': 0.07698991215957218, 'reg_lambda': 0.41594674297750694, 'min_child_samples': 11, 'num_leaves': 149}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:43:45,558] Trial 39 finished with value: 0.9267540233224727 and parameters: {'n_estimators': 337, 'max_depth': 7, 'learning_rate': 0.011309376724895317, 'subsample': 0.5728703419564883, 'colsample_bytree': 0.6861756981185496, 'reg_alpha': 0.9053039503750773, 'reg_lambda': 0.9924629503926156, 'min_child_samples': 18, 'num_leaves': 118}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:44:27,118] Trial 40 finished with value: 0.9281000099135621 and parameters: {'n_estimators': 276, 'max_depth': 6, 'learning_rate': 0.01746277757202098, 'subsample': 0.7626911046394872, 'colsample_bytree': 0.8521096133536548, 'reg_alpha': 0.43773253997123235, 'reg_lambda': 0.9144825733122011,

'min_child_samples': 8, 'num_leaves': 93}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:45:32,561] Trial 41 finished with value: 0.9285524419833727 and parameters: {'n_estimators': 418, 'max_depth': 7, 'learning_rate': 0.02131218364116074, 'subsample': 0.589452756033845, 'colsample_bytree': 0.8099067920553557, 'reg_alpha': 0.2689893788391492, 'reg_lambda': 0.7983556839997117, 'min_child_samples': 13, 'num_leaves': 76}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:46:45,173] Trial 42 finished with value: 0.9286344454819322 and parameters: {'n_estimators': 436, 'max_depth': 7, 'learning_rate': 0.027378903354994537, 'subsample': 0.653208922824731, 'colsample_bytree': 0.7669329435465286, 'reg_alpha': 0.14298419623420583, 'reg_lambda': 0.059804368030591006, 'min_child_samples': 24, 'num_leaves': 110}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:48:12,030] Trial 43 finished with value: 0.9286485840740684 and parameters: {'n_estimators': 467, 'max_depth': 8, 'learning_rate': 0.01587482879588693, 'subsample': 0.9486208002601322, 'colsample_bytree': 0.717441218469893, 'reg_alpha': 0.36423800730316663, 'reg_lambda': 0.5471028893076102, 'min_child_samples': 5, 'num_leaves': 127}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:49:04,709] Trial 44 finished with value: 0.9285439588600747 and parameters: {'n_estimators': 392, 'max_depth': 6, 'learning_rate': 0.04195336528309856, 'subsample': 0.8847764238824358, 'colsample_bytree': 0.8867207928872681, 'reg_alpha': 0.23726333026137272, 'reg_lambda': 0.06865427896199335, 'min_child_samples': 45, 'num_leaves': 104}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:49:59,658] Trial 45 finished with value: 0.9287023104283352 and parameters: {'n_estimators': 364, 'max_depth': 8, 'learning_rate': 0.020018931613123972, 'subsample': 0.6905616765913946, 'colsample_bytree': 0.8380320222799811, 'reg_alpha': 0.16739398940959158, 'reg_lambda': 0.28719370208799283, 'min_child_samples': 10, 'num_leaves': 69}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:50:14,139] Trial 46 finished with value: 0.9282018071532574 and parameters: {'n_estimators': 81, 'max_depth': 8, 'learning_rate': 0.011737897275271372, 'subsample': 0.7029802667712267, 'colsample_bytree': 0.844898018199468, 'reg_alpha': 0.170794237368529, 'reg_lambda': 0.43356298693870676, 'min_child_samples': 10, 'num_leaves': 83}. Best is trial 34 with value: 0.9287107933917136.

[I 2026-05-30 03:51:17,354] Trial 47 finished with value: 0.9286401010707099 and parameters: {'n_estimators': 366, 'max_depth': 8, 'learning_rate': 0.01914370990335157, 'subsample': 0.8228220415710978, 'colsample_bytree': 0.7999191238327891, 'reg_alpha': 0.05700109957332003, 'reg_lambda': 0.7130172287354305,

```
'min_child_samples': 8, 'num_leaves': 92}. Best is trial 34 with
value: 0.9287107933917136.
[I 2026-05-30 03:52:07,390] Trial 48 finished with value:
0.9287221042894105 and parameters: {'n_estimators': 342, 'max_depth':
8, 'learning_rate': 0.022455396702717036, 'subsample':
0.746560546696741, 'colsample_bytree': 0.8382372534588022,
'reg_alpha': 0.12470690485773948, 'reg_lambda': 0.27791225862561475,
'min_child_samples': 12, 'num_leaves': 70}. Best is trial 48 with
value: 0.9287221042894105.
[I 2026-05-30 03:52:56,085] Trial 49 finished with value:
0.9268529929476881 and parameters: {'n_estimators': 310, 'max_depth':
8, 'learning_rate': 0.021974521993992052, 'subsample':
0.68861364346442, 'colsample_bytree': 0.5437340001325845, 'reg_alpha':
0.11612582997117796, 'reg_lambda': 0.37851123857350993,
'min_child_samples': 13, 'num_leaves': 68}. Best is trial 48 with
value: 0.9287221042894105.
```

```
Best params: {'n_estimators': 342, 'max_depth': 8, 'learning_rate':
0.022455396702717036, 'subsample': 0.746560546696741,
'colsample_bytree': 0.8382372534588022, 'reg_alpha':
0.12470690485773948, 'reg_lambda': 0.27791225862561475,
'min_child_samples': 12, 'num_leaves': 70}
```

```
def objective_cat(trial):
    params = {
        'iterations' : trial.suggest_int('iterations', 50, 500),
        'depth'       : trial.suggest_int('depth', 2, 8),
        'learning_rate': trial.suggest_float('learning_rate', 0.01,
0.3, log=True),
        'l2_leaf_reg' : trial.suggest_float('l2_leaf_reg', 1.0, 10.0),
        'bagging_temperature':
trial.suggest_float('bagging_temperature', 0.0, 1.0),
        'random_strength' : trial.suggest_float('random_strength',
0.0, 1.0),
        'random_state': 42,
        'verbose'       : 0
    }
    model = CatBoostClassifier(**params)
    skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
    scores = cross_val_score(model, X_train, y_train, cv=skf,
scoring='accuracy')
    return scores.mean()
```

```
study_CATB = optuna.create_study(direction='maximize')
study_CATB.optimize(objective_cat, n_trials=50)
print("Best params:", study_CATB.best_params)
```

```
[I 2026-05-30 03:53:52,669] A new study created in memory with name:
no-name-bcf2b798-300d-4e4c-abae-7d994d0860e6
```

[I 2026-05-30 03:55:06,105] Trial 0 finished with value: 0.9263553177668482 and parameters: {'iterations': 255, 'depth': 6, 'learning_rate': 0.021997483888888412, 'l2_leaf_reg': 8.466706342449829, 'bagging_temperature': 0.13629606262308136, 'random_strength': 0.25731131361637105}. Best is trial 0 with value: 0.9263553177668482.

[I 2026-05-30 03:57:10,668] Trial 1 finished with value: 0.9287051380827945 and parameters: {'iterations': 408, 'depth': 7, 'learning_rate': 0.10547516223381109, 'l2_leaf_reg': 6.30248886932947, 'bagging_temperature': 0.5905631899962841, 'random_strength': 0.1707114033145034}. Best is trial 1 with value: 0.9287051380827945.

[I 2026-05-30 03:58:34,560] Trial 2 finished with value: 0.9281424262496888 and parameters: {'iterations': 317, 'depth': 5, 'learning_rate': 0.13372592882681292, 'l2_leaf_reg': 6.022302315952821, 'bagging_temperature': 0.4972807494916738, 'random_strength': 0.022618040089755653}. Best is trial 1 with value: 0.9287051380827945.

[I 2026-05-30 03:59:42,324] Trial 3 finished with value: 0.9260188221285668 and parameters: {'iterations': 236, 'depth': 6, 'learning_rate': 0.018714575950228976, 'l2_leaf_reg': 6.881636229743746, 'bagging_temperature': 0.016690841681820667, 'random_strength': 0.13630238622015767}. Best is trial 1 with value: 0.9287051380827945.

[I 2026-05-30 04:00:03,586] Trial 4 finished with value: 0.9196762843907642 and parameters: {'iterations': 173, 'depth': 3, 'learning_rate': 0.05338445740455266, 'l2_leaf_reg': 2.337658012155922, 'bagging_temperature': 0.09786137542423268, 'random_strength': 0.9892177163334424}. Best is trial 1 with value: 0.9287051380827945.

[I 2026-05-30 04:01:49,620] Trial 5 finished with value: 0.9285694086697467 and parameters: {'iterations': 342, 'depth': 7, 'learning_rate': 0.043499696230495905, 'l2_leaf_reg': 3.0737717406166594, 'bagging_temperature': 0.7380554376226298, 'random_strength': 0.532030366630638}. Best is trial 1 with value: 0.9287051380827945.

[I 2026-05-30 04:03:23,540] Trial 6 finished with value: 0.9286853442616991 and parameters: {'iterations': 306, 'depth': 7, 'learning_rate': 0.0578796972953409, 'l2_leaf_reg': 4.621204699976618, 'bagging_temperature': 0.510051987996551, 'random_strength': 0.5869813955625746}. Best is trial 1 with value: 0.9287051380827945.

[I 2026-05-30 04:04:50,008] Trial 7 finished with value: 0.9234456121927996 and parameters: {'iterations': 332, 'depth': 5, 'learning_rate': 0.011434029468625073, 'l2_leaf_reg': 4.837309684092223, 'bagging_temperature': 0.5613490807726895, 'random_strength': 0.47042453024298136}. Best is trial 1 with value: 0.9287051380827945.

[I 2026-05-30 04:04:59,385] Trial 8 finished with value: 0.9241779886107946 and parameters: {'iterations': 63, 'depth': 4, 'learning_rate': 0.2125714105436438, 'l2_leaf_reg': 2.160106151063356,

'bagging_temperature': 0.23207200565133945, 'random_strength': 0.9565761118538886}. Best is trial 1 with value: 0.9287051380827945.
[I 2026-05-30 04:07:03,514] Trial 9 finished with value: 0.9287221042894105 and parameters: {'iterations': 369, 'depth': 8, 'learning_rate': 0.03983517735615178, 'l2_leaf_reg': 9.946446490772223, 'bagging_temperature': 0.25654010098833024, 'random_strength': 0.2143162871043235}. Best is trial 9 with value: 0.9287221042894105.
[I 2026-05-30 04:09:44,175] Trial 10 finished with value: 0.9287192765949712 and parameters: {'iterations': 477, 'depth': 8, 'learning_rate': 0.026062339040270235, 'l2_leaf_reg': 9.88656443622529, 'bagging_temperature': 0.3186695626199083, 'random_strength': 0.360488051760481}. Best is trial 9 with value: 0.9287221042894105.
[I 2026-05-30 04:12:21,738] Trial 11 finished with value: 0.9287249319838498 and parameters: {'iterations': 472, 'depth': 8, 'learning_rate': 0.027557044728009655, 'l2_leaf_reg': 9.959247745148348, 'bagging_temperature': 0.28761889288250747, 'random_strength': 0.3599128078466545}. Best is trial 11 with value: 0.9287249319838498.
[I 2026-05-30 04:15:07,363] Trial 12 finished with value: 0.9285976856940998 and parameters: {'iterations': 492, 'depth': 8, 'learning_rate': 0.03559237263409823, 'l2_leaf_reg': 9.930661599515684, 'bagging_temperature': 0.31301547349014863, 'random_strength': 0.730384270943645}. Best is trial 11 with value: 0.9287249319838498.
[I 2026-05-30 04:16:29,717] Trial 13 finished with value: 0.9206886017986367 and parameters: {'iterations': 413, 'depth': 2, 'learning_rate': 0.07504598917125739, 'l2_leaf_reg': 7.964340233375041, 'bagging_temperature': 0.37279786180631813, 'random_strength': 0.319174528525672}. Best is trial 11 with value: 0.9287249319838498.
[I 2026-05-30 04:18:46,400] Trial 14 finished with value: 0.9284308507626624 and parameters: {'iterations': 413, 'depth': 8, 'learning_rate': 0.010669339289533947, 'l2_leaf_reg': 8.558344313134546, 'bagging_temperature': 0.8732493677470927, 'random_strength': 0.04212382830478581}. Best is trial 11 with value: 0.9287249319838498.
[I 2026-05-30 04:20:53,381] Trial 15 finished with value: 0.9279473144138379 and parameters: {'iterations': 447, 'depth': 6, 'learning_rate': 0.029605609943663024, 'l2_leaf_reg': 9.109903441002265, 'bagging_temperature': 0.1794164250245461, 'random_strength': 0.41598656071395856}. Best is trial 11 with value: 0.9287249319838498.
[I 2026-05-30 04:23:10,541] Trial 16 finished with value: 0.9287249319838498 and parameters: {'iterations': 407, 'depth': 8, 'learning_rate': 0.016619996583532703, 'l2_leaf_reg': 7.377958104880339, 'bagging_temperature': 0.41871515182365693, 'random_strength': 0.21182461318151605}. Best is trial 11 with value:

0.9287249319838498.
[I 2026-05-30 04:25:44,833] Trial 17 finished with value:
0.9283318817771244 and parameters: {'iterations': 500, 'depth': 7,
'learning_rate': 0.016338727011312372, 'l2_leaf_reg':
7.364801104240617, 'bagging_temperature': 0.41418400120473614,
'random_strength': 0.6520151841559624}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 04:26:09,004] Trial 18 finished with value:
0.919003291954786 and parameters: {'iterations': 187, 'depth': 4,
'learning_rate': 0.012889263987692511, 'l2_leaf_reg':
1.012837725082722, 'bagging_temperature': 0.6647183300176969,
'random_strength': 0.797621318428468}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 04:27:54,105] Trial 19 finished with value:
0.926442977293964 and parameters: {'iterations': 378, 'depth': 6,
'learning_rate': 0.017581133291781726, 'l2_leaf_reg':
5.407962792371007, 'bagging_temperature': 0.9747647189558108,
'random_strength': 0.30902999069090975}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 04:30:22,492] Trial 20 finished with value:
0.9287249319838498 and parameters: {'iterations': 456, 'depth': 8,
'learning_rate': 0.028178360575740594, 'l2_leaf_reg':
7.241015021780986, 'bagging_temperature': 0.4102160175020748,
'random_strength': 0.1159377008859909}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 04:32:47,155] Trial 21 finished with value:
0.9287249319838498 and parameters: {'iterations': 437, 'depth': 8,
'learning_rate': 0.028160231123279963, 'l2_leaf_reg':
7.347755104198063, 'bagging_temperature': 0.4288203675415435,
'random_strength': 0.12216169557007173}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 04:35:06,661] Trial 22 finished with value:
0.9286485836342901 and parameters: {'iterations': 459, 'depth': 7,
'learning_rate': 0.02271513976707214, 'l2_leaf_reg':
8.918151470578673, 'bagging_temperature': 0.45563790000956356,
'random_strength': 0.10598283367816957}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 04:37:14,118] Trial 23 finished with value:
0.9287136211661127 and parameters: {'iterations': 378, 'depth': 8,
'learning_rate': 0.015042186341800589, 'l2_leaf_reg':
7.525445204095138, 'bagging_temperature': 0.336759417452755,
'random_strength': 0.2486995154389104}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 04:39:31,860] Trial 24 finished with value:
0.9287164489005318 and parameters: {'iterations': 446, 'depth': 7,
'learning_rate': 0.03206937549754504, 'l2_leaf_reg':
6.567387542038181, 'bagging_temperature': 0.6698055403351579,
'random_strength': 0.3980529907668474}. Best is trial 11 with value:
0.9287249319838498.

[I 2026-05-30 04:41:47,750] Trial 25 finished with value: 0.9287249319838498 and parameters: {'iterations': 415, 'depth': 8, 'learning_rate': 0.05046452705067908, 'l2_leaf_reg': 3.8231538026439638, 'bagging_temperature': 0.252900110318888, 'random_strength': 0.20598019186373384}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:44:07,675] Trial 26 finished with value: 0.9286429287651492 and parameters: {'iterations': 468, 'depth': 7, 'learning_rate': 0.07010289182390152, 'l2_leaf_reg': 5.580285768023728, 'bagging_temperature': 0.3792496051000953, 'random_strength': 0.46259044860619003}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:44:19,850] Trial 27 finished with value: 0.9225831657086399 and parameters: {'iterations': 54, 'depth': 8, 'learning_rate': 0.020805870875712305, 'l2_leaf_reg': 8.084890241243349, 'bagging_temperature': 0.5730783187581199, 'random_strength': 0.30456347247112076}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:45:39,089] Trial 28 finished with value: 0.9262733146281074 and parameters: {'iterations': 285, 'depth': 6, 'learning_rate': 0.014058388417120022, 'l2_leaf_reg': 9.333165176408137, 'bagging_temperature': 0.004590784264839054, 'random_strength': 0.032389611610393235}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:47:04,266] Trial 29 finished with value: 0.9236661732786045 and parameters: {'iterations': 363, 'depth': 4, 'learning_rate': 0.020312944581423603, 'l2_leaf_reg': 8.171479960566952, 'bagging_temperature': 0.12338595416208903, 'random_strength': 0.2586438815826536}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:48:18,324] Trial 30 finished with value: 0.9283742968738757 and parameters: {'iterations': 245, 'depth': 7, 'learning_rate': 0.02200179595138724, 'l2_leaf_reg': 6.906754559020506, 'bagging_temperature': 0.1819006416426711, 'random_strength': 0.09230543415503402}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:50:40,728] Trial 31 finished with value: 0.9287249319838498 and parameters: {'iterations': 435, 'depth': 8, 'learning_rate': 0.02715985901557018, 'l2_leaf_reg': 7.35491468244362, 'bagging_temperature': 0.4075377345755472, 'random_strength': 0.1509188118941036}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:52:50,938] Trial 32 finished with value: 0.9287192765949712 and parameters: {'iterations': 396, 'depth': 8, 'learning_rate': 0.026235443995780237, 'l2_leaf_reg': 5.806811621906925, 'bagging_temperature': 0.4650852445179404, 'random_strength': 0.08478033722244796}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:55:05,978] Trial 33 finished with value: 0.9287051381227744 and parameters: {'iterations': 440, 'depth': 7,

'learning_rate': 0.03666158271300531, 'l2_leaf_reg': 8.612435067310848, 'bagging_temperature': 0.5259611480118547, 'random_strength': 0.1947884831154371}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:57:45,280] Trial 34 finished with value: 0.9286683778951638 and parameters: {'iterations': 481, 'depth': 8, 'learning_rate': 0.12763255742727297, 'l2_leaf_reg': 6.290992237158427, 'bagging_temperature': 0.4485093547574117, 'random_strength': 0.15166149967537734}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 04:59:55,885] Trial 35 finished with value: 0.928543958820095 and parameters: {'iterations': 429, 'depth': 7, 'learning_rate': 0.288380237995088, 'l2_leaf_reg': 6.933999901592467, 'bagging_temperature': 0.6466590936098563, 'random_strength': 0.2658670760899433}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 05:01:47,640] Trial 36 finished with value: 0.9281169761601579 and parameters: {'iterations': 397, 'depth': 6, 'learning_rate': 0.04195696257866178, 'l2_leaf_reg': 7.519183105944372, 'bagging_temperature': 0.2907025699455195, 'random_strength': 0.35618820725073363}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 05:04:20,220] Trial 37 finished with value: 0.9287221042894105 and parameters: {'iterations': 464, 'depth': 8, 'learning_rate': 0.030734134181137445, 'l2_leaf_reg': 4.810860087875272, 'bagging_temperature': 0.7648063162159586, 'random_strength': 0.06093221130770993}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 05:05:29,781] Trial 38 finished with value: 0.9163000159508385 and parameters: {'iterations': 347, 'depth': 2, 'learning_rate': 0.017842487999063232, 'l2_leaf_reg': 6.257534280295853, 'bagging_temperature': 0.36790955788677315, 'random_strength': 0.014899327347058428}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 05:06:51,002] Trial 39 finished with value: 0.9268699613132156 and parameters: {'iterations': 314, 'depth': 5, 'learning_rate': 0.06693024485124154, 'l2_leaf_reg': 9.39867108412037, 'bagging_temperature': 0.5193756044980256, 'random_strength': 0.1484435348639404}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 05:08:10,566] Trial 40 finished with value: 0.9280745613032856 and parameters: {'iterations': 281, 'depth': 6, 'learning_rate': 0.04757369165437904, 'l2_leaf_reg': 4.075070521575261, 'bagging_temperature': 0.2028267356232103, 'random_strength': 0.5241591270658194}. Best is trial 11 with value: 0.9287249319838498.

[I 2026-05-30 05:10:31,153] Trial 41 finished with value: 0.9287023103483752 and parameters: {'iterations': 424, 'depth': 8, 'learning_rate': 0.08301145973418213, 'l2_leaf_reg': 4.098292961217918, 'bagging_temperature': 0.08706402287526988, 'random_strength': 0.2039362913189029}. Best is trial 11 with value:

0.9287249319838498.
[I 2026-05-30 05:10:52,711] Trial 42 finished with value:
0.9282894657608367 and parameters: {'iterations': 101, 'depth': 8,
'learning_rate': 0.051279371976850424, 'l2_leaf_reg':
3.984536677505015, 'bagging_temperature': 0.23662971521709308,
'random_strength': 0.22251977864681904}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 05:13:08,033] Trial 43 finished with value:
0.9287249319838498 and parameters: {'iterations': 405, 'depth': 8,
'learning_rate': 0.023638597751807308, 'l2_leaf_reg':
2.6140339666726913, 'bagging_temperature': 0.28957403431753276,
'random_strength': 0.11743746645531265}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 05:15:35,562] Trial 44 finished with value:
0.9286683778951638 and parameters: {'iterations': 483, 'depth': 7,
'learning_rate': 0.10349412965013742, 'l2_leaf_reg':
3.496742586854204, 'bagging_temperature': 0.4270184852174964,
'random_strength': 0.17991974475132105}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 05:18:26,278] Trial 45 finished with value:
0.9287249319838498 and parameters: {'iterations': 500, 'depth': 8,
'learning_rate': 0.03519406267645171, 'l2_leaf_reg':
5.263239490118898, 'bagging_temperature': 0.3490757037218961,
'random_strength': 0.2828369272115545}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 05:20:45,355] Trial 46 finished with value:
0.9283290537628461 and parameters: {'iterations': 457, 'depth': 7,
'learning_rate': 0.012263475389250605, 'l2_leaf_reg':
1.6149784993014027, 'bagging_temperature': 0.059193629515699814,
'random_strength': 0.36773564484749216}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 05:21:32,577] Trial 47 finished with value:
0.9232448445682717 and parameters: {'iterations': 218, 'depth': 3,
'learning_rate': 0.055859208391633754, 'l2_leaf_reg':
7.745016871477656, 'bagging_temperature': 0.2819469957859243,
'random_strength': 0.23652103249801654}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 05:23:29,673] Trial 48 finished with value:
0.9286966549994766 and parameters: {'iterations': 347, 'depth': 8,
'learning_rate': 0.04433635428818758, 'l2_leaf_reg':
6.622135824748818, 'bagging_temperature': 0.49302183095343305,
'random_strength': 0.5733016065568879}. Best is trial 11 with value:
0.9287249319838498.
[I 2026-05-30 05:25:40,842] Trial 49 finished with value:
0.9287192765949712 and parameters: {'iterations': 391, 'depth': 8,
'learning_rate': 0.025800798958355766, 'l2_leaf_reg':
3.1425470154297503, 'bagging_temperature': 0.24303680541642558,
'random_strength': 0.06968126434173526}. Best is trial 11 with value:
0.9287249319838498.

```
Best params: {'iterations': 472, 'depth': 8, 'learning_rate':  
0.027557044728009655, 'l2_leaf_reg': 9.959247745148348,  
'bagging_temperature': 0.28761889288250747, 'random_strength':  
0.3599128078466545}
```

I will now train the 3 models on full data

XGBoost

```
xgb_model = XGBClassifier(  
    **study_XGB.best_params,  
    eval_metric='logloss'  
)  
  
xgb_model.fit(X_train, y_train)  
xgb_pred = xgb_model.predict(X_test)
```

LightGBM

```
lgbm_model = LGBMClassifier(  
    **study_LGBM.best_params,  
    verbose=-1  
)  
  
lgbm_model.fit(X_train, y_train)  
lgbm_pred = lgbm_model.predict(X_test)
```

CatBoost

```
cat_model = CatBoostClassifier(  
    **study_CATB.best_params  
)  
  
cat_model.fit(X_train, y_train)  
cat_pred = cat_model.predict(X_test)
```

```
0:   learn: 0.6617057 total: 69.9ms   remaining: 32.9s  
1:   learn: 0.6331515 total: 141ms   remaining: 33s  
2:   learn: 0.6051289 total: 210ms   remaining: 32.9s  
3:   learn: 0.5793447 total: 299ms   remaining: 35s  
4:   learn: 0.5554366 total: 369ms   remaining: 34.5s  
5:   learn: 0.5340876 total: 438ms   remaining: 34s  
6:   learn: 0.5146451 total: 509ms   remaining: 33.8s  
7:   learn: 0.4955581 total: 578ms   remaining: 33.5s  
8:   learn: 0.4788850 total: 649ms   remaining: 33.4s  
9:   learn: 0.4630717 total: 718ms   remaining: 33.2s  
10:  learn: 0.4482922 total: 787ms   remaining: 33s  
11:  learn: 0.4347536 total: 857ms   remaining: 32.9s  
12:  learn: 0.4220724 total: 930ms   remaining: 32.9s  
13:  learn: 0.4095588 total: 1s      remaining: 32.8s
```

14:	learn: 0.3990171	total: 1.07s	remaining: 32.7s
15:	learn: 0.3892562	total: 1.15s	remaining: 32.7s
16:	learn: 0.3800379	total: 1.22s	remaining: 32.6s
17:	learn: 0.3716667	total: 1.31s	remaining: 33s
18:	learn: 0.3637043	total: 1.39s	remaining: 33s
19:	learn: 0.3564898	total: 1.46s	remaining: 33s
20:	learn: 0.3497829	total: 1.53s	remaining: 32.9s
21:	learn: 0.3426558	total: 1.6s	remaining: 32.8s
22:	learn: 0.3360618	total: 1.67s	remaining: 32.7s
23:	learn: 0.3306482	total: 1.75s	remaining: 32.6s
24:	learn: 0.3248759	total: 1.82s	remaining: 32.5s
25:	learn: 0.3196202	total: 1.89s	remaining: 32.4s
26:	learn: 0.3147952	total: 1.96s	remaining: 32.4s
27:	learn: 0.3102813	total: 2.04s	remaining: 32.3s
28:	learn: 0.3061248	total: 2.11s	remaining: 32.3s
29:	learn: 0.3021838	total: 2.19s	remaining: 32.2s
30:	learn: 0.2985216	total: 2.26s	remaining: 32.2s
31:	learn: 0.2951483	total: 2.35s	remaining: 32.3s
32:	learn: 0.2919196	total: 2.42s	remaining: 32.3s
33:	learn: 0.2890500	total: 2.5s	remaining: 32.2s
34:	learn: 0.2863447	total: 2.57s	remaining: 32.1s
35:	learn: 0.2836903	total: 2.64s	remaining: 32s
36:	learn: 0.2813100	total: 2.72s	remaining: 32s
37:	learn: 0.2791162	total: 2.79s	remaining: 31.9s
38:	learn: 0.2770007	total: 2.86s	remaining: 31.8s
39:	learn: 0.2750191	total: 2.94s	remaining: 31.7s
40:	learn: 0.2732311	total: 3.01s	remaining: 31.6s
41:	learn: 0.2714616	total: 3.1s	remaining: 31.7s
42:	learn: 0.2697804	total: 3.17s	remaining: 31.6s
43:	learn: 0.2682722	total: 3.24s	remaining: 31.6s
44:	learn: 0.2668150	total: 3.31s	remaining: 31.5s
45:	learn: 0.2655476	total: 3.41s	remaining: 31.6s
46:	learn: 0.2643102	total: 3.53s	remaining: 31.9s
47:	learn: 0.2631723	total: 3.65s	remaining: 32.3s
48:	learn: 0.2621200	total: 3.78s	remaining: 32.6s
49:	learn: 0.2610765	total: 3.91s	remaining: 33s
50:	learn: 0.2600992	total: 4.02s	remaining: 33.2s
51:	learn: 0.2591942	total: 4.16s	remaining: 33.6s
52:	learn: 0.2582972	total: 4.28s	remaining: 33.8s
53:	learn: 0.2575158	total: 4.4s	remaining: 34.1s
54:	learn: 0.2567260	total: 4.52s	remaining: 34.3s
55:	learn: 0.2560523	total: 4.64s	remaining: 34.5s
56:	learn: 0.2553765	total: 4.77s	remaining: 34.7s
57:	learn: 0.2547174	total: 4.89s	remaining: 34.9s
58:	learn: 0.2541192	total: 5.01s	remaining: 35.1s
59:	learn: 0.2535288	total: 5.12s	remaining: 35.2s
60:	learn: 0.2527455	total: 5.25s	remaining: 35.4s
61:	learn: 0.2519922	total: 5.37s	remaining: 35.5s
62:	learn: 0.2512869	total: 5.49s	remaining: 35.7s

63:	learn: 0.2508685	total: 5.61s	remaining: 35.8s
64:	learn: 0.2502030	total: 5.74s	remaining: 35.9s
65:	learn: 0.2495979	total: 5.86s	remaining: 36.1s
66:	learn: 0.2489945	total: 5.99s	remaining: 36.2s
67:	learn: 0.2484507	total: 6.11s	remaining: 36.3s
68:	learn: 0.2479213	total: 6.23s	remaining: 36.4s
69:	learn: 0.2474417	total: 6.35s	remaining: 36.5s
70:	learn: 0.2469873	total: 6.47s	remaining: 36.5s
71:	learn: 0.2465736	total: 6.59s	remaining: 36.6s
72:	learn: 0.2461391	total: 6.7s	remaining: 36.6s
73:	learn: 0.2457268	total: 6.83s	remaining: 36.7s
74:	learn: 0.2453865	total: 6.95s	remaining: 36.8s
75:	learn: 0.2450216	total: 7.07s	remaining: 36.8s
76:	learn: 0.2446604	total: 7.21s	remaining: 37s
77:	learn: 0.2443574	total: 7.35s	remaining: 37.1s
78:	learn: 0.2440829	total: 7.48s	remaining: 37.2s
79:	learn: 0.2438363	total: 7.62s	remaining: 37.3s
80:	learn: 0.2435400	total: 7.75s	remaining: 37.4s
81:	learn: 0.2433433	total: 7.87s	remaining: 37.4s
82:	learn: 0.2430732	total: 7.97s	remaining: 37.4s
83:	learn: 0.2428483	total: 8.04s	remaining: 37.1s
84:	learn: 0.2426367	total: 8.11s	remaining: 36.9s
85:	learn: 0.2424254	total: 8.19s	remaining: 36.7s
86:	learn: 0.2422298	total: 8.26s	remaining: 36.5s
87:	learn: 0.2420525	total: 8.33s	remaining: 36.4s
88:	learn: 0.2419071	total: 8.41s	remaining: 36.2s
89:	learn: 0.2417747	total: 8.48s	remaining: 36s
90:	learn: 0.2416342	total: 8.55s	remaining: 35.8s
91:	learn: 0.2414768	total: 8.63s	remaining: 35.6s
92:	learn: 0.2413111	total: 8.7s	remaining: 35.4s
93:	learn: 0.2411562	total: 8.79s	remaining: 35.3s
94:	learn: 0.2410221	total: 8.86s	remaining: 35.2s
95:	learn: 0.2408836	total: 8.94s	remaining: 35s
96:	learn: 0.2407647	total: 9.01s	remaining: 34.8s
97:	learn: 0.2406628	total: 9.08s	remaining: 34.7s
98:	learn: 0.2405097	total: 9.16s	remaining: 34.5s
99:	learn: 0.2403869	total: 9.23s	remaining: 34.3s
100:	learn: 0.2402536	total: 9.31s	remaining: 34.2s
101:	learn: 0.2401768	total: 9.38s	remaining: 34s
102:	learn: 0.2400689	total: 9.46s	remaining: 33.9s
103:	learn: 0.2399948	total: 9.53s	remaining: 33.7s
104:	learn: 0.2399059	total: 9.6s	remaining: 33.5s
105:	learn: 0.2398042	total: 9.67s	remaining: 33.4s
106:	learn: 0.2396666	total: 9.74s	remaining: 33.2s
107:	learn: 0.2395885	total: 9.82s	remaining: 33.1s
108:	learn: 0.2394944	total: 9.9s	remaining: 33s
109:	learn: 0.2393929	total: 9.97s	remaining: 32.8s
110:	learn: 0.2393054	total: 10s	remaining: 32.7s
111:	learn: 0.2392345	total: 10.1s	remaining: 32.5s

112:	learn:	0.2391453	total:	10.2s	remaining:	32.4s
113:	learn:	0.2390740	total:	10.3s	remaining:	32.2s
114:	learn:	0.2390132	total:	10.3s	remaining:	32.1s
115:	learn:	0.2389153	total:	10.4s	remaining:	32s
116:	learn:	0.2388260	total:	10.5s	remaining:	31.8s
117:	learn:	0.2387673	total:	10.6s	remaining:	31.7s
118:	learn:	0.2386693	total:	10.6s	remaining:	31.5s
119:	learn:	0.2385844	total:	10.7s	remaining:	31.4s
120:	learn:	0.2385211	total:	10.8s	remaining:	31.3s
121:	learn:	0.2384579	total:	10.9s	remaining:	31.2s
122:	learn:	0.2383849	total:	10.9s	remaining:	31s
123:	learn:	0.2383220	total:	11s	remaining:	30.9s
124:	learn:	0.2382556	total:	11.1s	remaining:	30.8s
125:	learn:	0.2382098	total:	11.2s	remaining:	30.7s
126:	learn:	0.2381575	total:	11.2s	remaining:	30.5s
127:	learn:	0.2381056	total:	11.3s	remaining:	30.4s
128:	learn:	0.2380698	total:	11.4s	remaining:	30.3s
129:	learn:	0.2380367	total:	11.5s	remaining:	30.1s
130:	learn:	0.2379904	total:	11.5s	remaining:	30s
131:	learn:	0.2379302	total:	11.6s	remaining:	29.9s
132:	learn:	0.2378861	total:	11.7s	remaining:	29.7s
133:	learn:	0.2378529	total:	11.7s	remaining:	29.6s
134:	learn:	0.2378053	total:	11.8s	remaining:	29.5s
135:	learn:	0.2377540	total:	11.9s	remaining:	29.4s
136:	learn:	0.2377256	total:	12s	remaining:	29.3s
137:	learn:	0.2376803	total:	12.1s	remaining:	29.2s
138:	learn:	0.2376263	total:	12.1s	remaining:	29s
139:	learn:	0.2376102	total:	12.2s	remaining:	28.9s
140:	learn:	0.2375748	total:	12.3s	remaining:	28.8s
141:	learn:	0.2375587	total:	12.3s	remaining:	28.7s
142:	learn:	0.2375403	total:	12.4s	remaining:	28.6s
143:	learn:	0.2375224	total:	12.5s	remaining:	28.5s
144:	learn:	0.2374767	total:	12.6s	remaining:	28.3s
145:	learn:	0.2374373	total:	12.6s	remaining:	28.2s
146:	learn:	0.2374018	total:	12.7s	remaining:	28.1s
147:	learn:	0.2373835	total:	12.8s	remaining:	28s
148:	learn:	0.2373682	total:	12.9s	remaining:	27.9s
149:	learn:	0.2373500	total:	13s	remaining:	27.8s
150:	learn:	0.2373052	total:	13s	remaining:	27.7s
151:	learn:	0.2372891	total:	13.1s	remaining:	27.6s
152:	learn:	0.2372672	total:	13.2s	remaining:	27.5s
153:	learn:	0.2372365	total:	13.3s	remaining:	27.4s
154:	learn:	0.2372065	total:	13.4s	remaining:	27.3s
155:	learn:	0.2371925	total:	13.4s	remaining:	27.2s
156:	learn:	0.2371616	total:	13.5s	remaining:	27.1s
157:	learn:	0.2371392	total:	13.6s	remaining:	27s
158:	learn:	0.2371242	total:	13.6s	remaining:	26.9s
159:	learn:	0.2371091	total:	13.7s	remaining:	26.8s
160:	learn:	0.2370835	total:	13.8s	remaining:	26.7s

161: learn: 0.2370592 total: 13.9s remaining: 26.5s
162: learn: 0.2370437 total: 14s remaining: 26.5s
163: learn: 0.2370150 total: 14s remaining: 26.4s
164: learn: 0.2369929 total: 14.1s remaining: 26.2s
165: learn: 0.2369741 total: 14.2s remaining: 26.1s
166: learn: 0.2369521 total: 14.3s remaining: 26s
167: learn: 0.2369291 total: 14.3s remaining: 25.9s
168: learn: 0.2369048 total: 14.4s remaining: 25.8s
169: learn: 0.2368794 total: 14.5s remaining: 25.7s
170: learn: 0.2368591 total: 14.5s remaining: 25.6s
171: learn: 0.2368437 total: 14.6s remaining: 25.5s
172: learn: 0.2368283 total: 14.7s remaining: 25.4s
173: learn: 0.2368067 total: 14.8s remaining: 25.3s
174: learn: 0.2367953 total: 14.8s remaining: 25.2s
175: learn: 0.2367836 total: 14.9s remaining: 25.1s
176: learn: 0.2367714 total: 15s remaining: 25s
177: learn: 0.2367572 total: 15.1s remaining: 24.9s
178: learn: 0.2367483 total: 15.1s remaining: 24.8s
179: learn: 0.2367393 total: 15.2s remaining: 24.7s
180: learn: 0.2367140 total: 15.3s remaining: 24.6s
181: learn: 0.2367030 total: 15.4s remaining: 24.5s
182: learn: 0.2366873 total: 15.4s remaining: 24.4s
183: learn: 0.2366771 total: 15.5s remaining: 24.3s
184: learn: 0.2366618 total: 15.6s remaining: 24.2s
185: learn: 0.2366425 total: 15.7s remaining: 24.1s
186: learn: 0.2366222 total: 15.7s remaining: 24s
187: learn: 0.2366118 total: 15.8s remaining: 23.9s
188: learn: 0.2366015 total: 15.9s remaining: 23.8s
189: learn: 0.2365904 total: 16s remaining: 23.7s
190: learn: 0.2365827 total: 16s remaining: 23.6s
191: learn: 0.2365731 total: 16.1s remaining: 23.5s
192: learn: 0.2365602 total: 16.2s remaining: 23.4s
193: learn: 0.2365500 total: 16.3s remaining: 23.3s
194: learn: 0.2365387 total: 16.3s remaining: 23.2s
195: learn: 0.2365300 total: 16.4s remaining: 23.1s
196: learn: 0.2365160 total: 16.5s remaining: 23s
197: learn: 0.2365064 total: 16.6s remaining: 22.9s
198: learn: 0.2364966 total: 16.6s remaining: 22.8s
199: learn: 0.2364889 total: 16.7s remaining: 22.7s
200: learn: 0.2364822 total: 16.8s remaining: 22.6s
201: learn: 0.2364728 total: 16.8s remaining: 22.5s
202: learn: 0.2364585 total: 16.9s remaining: 22.4s
203: learn: 0.2364351 total: 17s remaining: 22.3s
204: learn: 0.2364231 total: 17.1s remaining: 22.2s
205: learn: 0.2364117 total: 17.1s remaining: 22.1s
206: learn: 0.2364036 total: 17.2s remaining: 22s
207: learn: 0.2363854 total: 17.3s remaining: 21.9s
208: learn: 0.2363754 total: 17.4s remaining: 21.9s
209: learn: 0.2363658 total: 17.4s remaining: 21.8s

210:	learn: 0.2363569	total: 17.5s	remaining: 21.7s
211:	learn: 0.2363391	total: 17.6s	remaining: 21.6s
212:	learn: 0.2363228	total: 17.7s	remaining: 21.5s
213:	learn: 0.2363140	total: 17.7s	remaining: 21.4s
214:	learn: 0.2363004	total: 17.8s	remaining: 21.3s
215:	learn: 0.2362888	total: 17.9s	remaining: 21.2s
216:	learn: 0.2362802	total: 17.9s	remaining: 21.1s
217:	learn: 0.2362712	total: 18.1s	remaining: 21.1s
218:	learn: 0.2362647	total: 18.2s	remaining: 21s
219:	learn: 0.2362560	total: 18.3s	remaining: 21s
220:	learn: 0.2362460	total: 18.4s	remaining: 20.9s
221:	learn: 0.2362359	total: 18.6s	remaining: 20.9s
222:	learn: 0.2362265	total: 18.7s	remaining: 20.9s
223:	learn: 0.2362140	total: 18.8s	remaining: 20.8s
224:	learn: 0.2362034	total: 18.9s	remaining: 20.8s
225:	learn: 0.2361946	total: 19.1s	remaining: 20.7s
226:	learn: 0.2361865	total: 19.2s	remaining: 20.7s
227:	learn: 0.2361792	total: 19.3s	remaining: 20.7s
228:	learn: 0.2361704	total: 19.4s	remaining: 20.6s
229:	learn: 0.2361652	total: 19.5s	remaining: 20.6s
230:	learn: 0.2361558	total: 19.7s	remaining: 20.5s
231:	learn: 0.2361427	total: 19.8s	remaining: 20.5s
232:	learn: 0.2361318	total: 19.9s	remaining: 20.4s
233:	learn: 0.2361184	total: 20s	remaining: 20.4s
234:	learn: 0.2361111	total: 20.1s	remaining: 20.3s
235:	learn: 0.2360998	total: 20.3s	remaining: 20.3s
236:	learn: 0.2360916	total: 20.4s	remaining: 20.2s
237:	learn: 0.2360831	total: 20.5s	remaining: 20.2s
238:	learn: 0.2360746	total: 20.7s	remaining: 20.1s
239:	learn: 0.2360648	total: 20.8s	remaining: 20.1s
240:	learn: 0.2360566	total: 20.9s	remaining: 20s
241:	learn: 0.2360472	total: 21s	remaining: 20s
242:	learn: 0.2360383	total: 21.1s	remaining: 19.9s
243:	learn: 0.2360303	total: 21.2s	remaining: 19.8s
244:	learn: 0.2360187	total: 21.4s	remaining: 19.8s
245:	learn: 0.2359908	total: 21.5s	remaining: 19.7s
246:	learn: 0.2359806	total: 21.6s	remaining: 19.7s
247:	learn: 0.2359728	total: 21.7s	remaining: 19.6s
248:	learn: 0.2359652	total: 21.8s	remaining: 19.6s
249:	learn: 0.2359518	total: 22s	remaining: 19.5s
250:	learn: 0.2359437	total: 22.1s	remaining: 19.4s
251:	learn: 0.2359350	total: 22.2s	remaining: 19.4s
252:	learn: 0.2359251	total: 22.3s	remaining: 19.3s
253:	learn: 0.2359185	total: 22.4s	remaining: 19.2s
254:	learn: 0.2359129	total: 22.5s	remaining: 19.1s
255:	learn: 0.2359042	total: 22.6s	remaining: 19s
256:	learn: 0.2358935	total: 22.6s	remaining: 18.9s
257:	learn: 0.2358865	total: 22.7s	remaining: 18.8s
258:	learn: 0.2358776	total: 22.8s	remaining: 18.7s

259: learn: 0.2358691 total: 22.8s remaining: 18.6s
260: learn: 0.2358579 total: 22.9s remaining: 18.5s
261: learn: 0.2358475 total: 23s remaining: 18.4s
262: learn: 0.2358390 total: 23.1s remaining: 18.3s
263: learn: 0.2358334 total: 23.1s remaining: 18.2s
264: learn: 0.2358243 total: 23.2s remaining: 18.1s
265: learn: 0.2358167 total: 23.3s remaining: 18s
266: learn: 0.2358087 total: 23.3s remaining: 17.9s
267: learn: 0.2358016 total: 23.4s remaining: 17.8s
268: learn: 0.2357958 total: 23.5s remaining: 17.7s
269: learn: 0.2357869 total: 23.6s remaining: 17.6s
270: learn: 0.2357782 total: 23.7s remaining: 17.5s
271: learn: 0.2357690 total: 23.7s remaining: 17.4s
272: learn: 0.2357595 total: 23.8s remaining: 17.3s
273: learn: 0.2357505 total: 23.9s remaining: 17.2s
274: learn: 0.2357439 total: 23.9s remaining: 17.1s
275: learn: 0.2357378 total: 24s remaining: 17s
276: learn: 0.2357305 total: 24.1s remaining: 17s
277: learn: 0.2357245 total: 24.1s remaining: 16.9s
278: learn: 0.2357161 total: 24.2s remaining: 16.8s
279: learn: 0.2357107 total: 24.3s remaining: 16.7s
280: learn: 0.2357036 total: 24.4s remaining: 16.6s
281: learn: 0.2356964 total: 24.4s remaining: 16.5s
282: learn: 0.2356856 total: 24.5s remaining: 16.4s
283: learn: 0.2356777 total: 24.6s remaining: 16.3s
284: learn: 0.2356688 total: 24.6s remaining: 16.2s
285: learn: 0.2356597 total: 24.7s remaining: 16.1s
286: learn: 0.2356528 total: 24.8s remaining: 16s
287: learn: 0.2356445 total: 24.9s remaining: 15.9s
288: learn: 0.2356267 total: 24.9s remaining: 15.8s
289: learn: 0.2356207 total: 25s remaining: 15.7s
290: learn: 0.2356122 total: 25.1s remaining: 15.6s
291: learn: 0.2356045 total: 25.1s remaining: 15.5s
292: learn: 0.2355981 total: 25.2s remaining: 15.4s
293: learn: 0.2355916 total: 25.3s remaining: 15.3s
294: learn: 0.2355826 total: 25.4s remaining: 15.2s
295: learn: 0.2355751 total: 25.4s remaining: 15.1s
296: learn: 0.2355679 total: 25.5s remaining: 15s
297: learn: 0.2355623 total: 25.6s remaining: 14.9s
298: learn: 0.2355560 total: 25.7s remaining: 14.9s
299: learn: 0.2355497 total: 25.7s remaining: 14.8s
300: learn: 0.2355416 total: 25.8s remaining: 14.7s
301: learn: 0.2355323 total: 25.9s remaining: 14.6s
302: learn: 0.2355240 total: 26s remaining: 14.5s
303: learn: 0.2355174 total: 26s remaining: 14.4s
304: learn: 0.2355114 total: 26.1s remaining: 14.3s
305: learn: 0.2355046 total: 26.2s remaining: 14.2s
306: learn: 0.2354975 total: 26.2s remaining: 14.1s
307: learn: 0.2354908 total: 26.3s remaining: 14s

308:	learn:	0.2354768	total:	26.4s	remaining:	13.9s
309:	learn:	0.2354694	total:	26.5s	remaining:	13.8s
310:	learn:	0.2354549	total:	26.5s	remaining:	13.7s
311:	learn:	0.2354515	total:	26.6s	remaining:	13.6s
312:	learn:	0.2354427	total:	26.7s	remaining:	13.5s
313:	learn:	0.2354363	total:	26.7s	remaining:	13.5s
314:	learn:	0.2354273	total:	26.8s	remaining:	13.4s
315:	learn:	0.2354213	total:	26.9s	remaining:	13.3s
316:	learn:	0.2354045	total:	27s	remaining:	13.2s
317:	learn:	0.2353941	total:	27s	remaining:	13.1s
318:	learn:	0.2353884	total:	27.1s	remaining:	13s
319:	learn:	0.2353794	total:	27.2s	remaining:	12.9s
320:	learn:	0.2353701	total:	27.2s	remaining:	12.8s
321:	learn:	0.2353591	total:	27.3s	remaining:	12.7s
322:	learn:	0.2353527	total:	27.4s	remaining:	12.6s
323:	learn:	0.2353446	total:	27.5s	remaining:	12.5s
324:	learn:	0.2353378	total:	27.5s	remaining:	12.5s
325:	learn:	0.2353272	total:	27.6s	remaining:	12.4s
326:	learn:	0.2353207	total:	27.7s	remaining:	12.3s
327:	learn:	0.2353114	total:	27.8s	remaining:	12.2s
328:	learn:	0.2353028	total:	27.9s	remaining:	12.1s
329:	learn:	0.2352932	total:	27.9s	remaining:	12s
330:	learn:	0.2352866	total:	28s	remaining:	11.9s
331:	learn:	0.2352789	total:	28.1s	remaining:	11.8s
332:	learn:	0.2352650	total:	28.1s	remaining:	11.7s
333:	learn:	0.2352543	total:	28.2s	remaining:	11.7s
334:	learn:	0.2352480	total:	28.3s	remaining:	11.6s
335:	learn:	0.2352396	total:	28.4s	remaining:	11.5s
336:	learn:	0.2352323	total:	28.4s	remaining:	11.4s
337:	learn:	0.2352267	total:	28.5s	remaining:	11.3s
338:	learn:	0.2352216	total:	28.6s	remaining:	11.2s
339:	learn:	0.2352136	total:	28.7s	remaining:	11.1s
340:	learn:	0.2352028	total:	28.8s	remaining:	11s
341:	learn:	0.2351936	total:	28.8s	remaining:	11s
342:	learn:	0.2351861	total:	28.9s	remaining:	10.9s
343:	learn:	0.2351791	total:	29s	remaining:	10.8s
344:	learn:	0.2351703	total:	29.1s	remaining:	10.7s
345:	learn:	0.2351606	total:	29.1s	remaining:	10.6s
346:	learn:	0.2351525	total:	29.2s	remaining:	10.5s
347:	learn:	0.2351474	total:	29.3s	remaining:	10.4s
348:	learn:	0.2351406	total:	29.3s	remaining:	10.3s
349:	learn:	0.2351363	total:	29.4s	remaining:	10.3s
350:	learn:	0.2351286	total:	29.5s	remaining:	10.2s
351:	learn:	0.2351195	total:	29.6s	remaining:	10.1s
352:	learn:	0.2351135	total:	29.6s	remaining:	9.99s
353:	learn:	0.2351050	total:	29.7s	remaining:	9.9s
354:	learn:	0.2350962	total:	29.8s	remaining:	9.82s
355:	learn:	0.2350888	total:	29.9s	remaining:	9.73s
356:	learn:	0.2350812	total:	29.9s	remaining:	9.64s

357: learn: 0.2350734 total: 30s remaining: 9.55s
358: learn: 0.2350677 total: 30.1s remaining: 9.47s
359: learn: 0.2350597 total: 30.1s remaining: 9.38s
360: learn: 0.2350516 total: 30.2s remaining: 9.29s
361: learn: 0.2350374 total: 30.3s remaining: 9.21s
362: learn: 0.2350300 total: 30.4s remaining: 9.12s
363: learn: 0.2350235 total: 30.5s remaining: 9.04s
364: learn: 0.2350164 total: 30.5s remaining: 8.95s
365: learn: 0.2350096 total: 30.6s remaining: 8.86s
366: learn: 0.2350009 total: 30.7s remaining: 8.77s
367: learn: 0.2349929 total: 30.7s remaining: 8.68s
368: learn: 0.2349867 total: 30.8s remaining: 8.6s
369: learn: 0.2349783 total: 30.9s remaining: 8.52s
370: learn: 0.2349732 total: 31s remaining: 8.43s
371: learn: 0.2349650 total: 31s remaining: 8.34s
372: learn: 0.2349587 total: 31.1s remaining: 8.26s
373: learn: 0.2349513 total: 31.2s remaining: 8.17s
374: learn: 0.2349432 total: 31.3s remaining: 8.08s
375: learn: 0.2349369 total: 31.3s remaining: 8s
376: learn: 0.2349295 total: 31.4s remaining: 7.91s
377: learn: 0.2349237 total: 31.5s remaining: 7.82s
378: learn: 0.2349180 total: 31.5s remaining: 7.74s
379: learn: 0.2349084 total: 31.6s remaining: 7.65s
380: learn: 0.2348948 total: 31.7s remaining: 7.57s
381: learn: 0.2348900 total: 31.8s remaining: 7.48s
382: learn: 0.2348826 total: 31.8s remaining: 7.4s
383: learn: 0.2348743 total: 31.9s remaining: 7.31s
384: learn: 0.2348642 total: 32s remaining: 7.23s
385: learn: 0.2348565 total: 32.1s remaining: 7.14s
386: learn: 0.2348473 total: 32.1s remaining: 7.06s
387: learn: 0.2348387 total: 32.2s remaining: 6.97s
388: learn: 0.2348300 total: 32.3s remaining: 6.89s
389: learn: 0.2348239 total: 32.4s remaining: 6.81s
390: learn: 0.2348169 total: 32.5s remaining: 6.74s
391: learn: 0.2348096 total: 32.7s remaining: 6.67s
392: learn: 0.2348049 total: 32.8s remaining: 6.59s
393: learn: 0.2347978 total: 32.9s remaining: 6.52s
394: learn: 0.2347897 total: 33.1s remaining: 6.44s
395: learn: 0.2347836 total: 33.2s remaining: 6.37s
396: learn: 0.2347764 total: 33.3s remaining: 6.29s
397: learn: 0.2347664 total: 33.4s remaining: 6.21s
398: learn: 0.2347595 total: 33.6s remaining: 6.14s
399: learn: 0.2347519 total: 33.7s remaining: 6.06s
400: learn: 0.2347446 total: 33.8s remaining: 5.98s
401: learn: 0.2347407 total: 33.9s remaining: 5.91s
402: learn: 0.2347342 total: 34s remaining: 5.83s
403: learn: 0.2347299 total: 34.2s remaining: 5.75s
404: learn: 0.2347248 total: 34.3s remaining: 5.67s
405: learn: 0.2347200 total: 34.4s remaining: 5.59s

406:	learn:	0.2347151	total:	34.5s	remaining:	5.51s
407:	learn:	0.2347056	total:	34.6s	remaining:	5.43s
408:	learn:	0.2347004	total:	34.8s	remaining:	5.36s
409:	learn:	0.2346946	total:	34.9s	remaining:	5.28s
410:	learn:	0.2346888	total:	35s	remaining:	5.2s
411:	learn:	0.2346782	total:	35.1s	remaining:	5.12s
412:	learn:	0.2346731	total:	35.3s	remaining:	5.04s
413:	learn:	0.2346662	total:	35.4s	remaining:	4.96s
414:	learn:	0.2346565	total:	35.5s	remaining:	4.88s
415:	learn:	0.2346486	total:	35.6s	remaining:	4.8s
416:	learn:	0.2346433	total:	35.8s	remaining:	4.71s
417:	learn:	0.2346334	total:	35.9s	remaining:	4.63s
418:	learn:	0.2346279	total:	36s	remaining:	4.55s
419:	learn:	0.2346234	total:	36.1s	remaining:	4.47s
420:	learn:	0.2346171	total:	36.3s	remaining:	4.39s
421:	learn:	0.2346099	total:	36.4s	remaining:	4.31s
422:	learn:	0.2346029	total:	36.5s	remaining:	4.23s
423:	learn:	0.2345957	total:	36.6s	remaining:	4.15s
424:	learn:	0.2345902	total:	36.7s	remaining:	4.06s
425:	learn:	0.2345818	total:	36.8s	remaining:	3.97s
426:	learn:	0.2345737	total:	36.8s	remaining:	3.88s
427:	learn:	0.2345687	total:	36.9s	remaining:	3.79s
428:	learn:	0.2345604	total:	37s	remaining:	3.71s
429:	learn:	0.2345559	total:	37s	remaining:	3.62s
430:	learn:	0.2345512	total:	37.1s	remaining:	3.53s
431:	learn:	0.2345466	total:	37.2s	remaining:	3.44s
432:	learn:	0.2345403	total:	37.3s	remaining:	3.36s
433:	learn:	0.2345341	total:	37.3s	remaining:	3.27s
434:	learn:	0.2345208	total:	37.4s	remaining:	3.18s
435:	learn:	0.2345106	total:	37.5s	remaining:	3.1s
436:	learn:	0.2345022	total:	37.6s	remaining:	3.01s
437:	learn:	0.2344943	total:	37.6s	remaining:	2.92s
438:	learn:	0.2344859	total:	37.7s	remaining:	2.83s
439:	learn:	0.2344768	total:	37.8s	remaining:	2.75s
440:	learn:	0.2344712	total:	37.9s	remaining:	2.66s
441:	learn:	0.2344621	total:	37.9s	remaining:	2.58s
442:	learn:	0.2344541	total:	38s	remaining:	2.49s
443:	learn:	0.2344493	total:	38.1s	remaining:	2.4s
444:	learn:	0.2344425	total:	38.2s	remaining:	2.31s
445:	learn:	0.2344336	total:	38.2s	remaining:	2.23s
446:	learn:	0.2344271	total:	38.3s	remaining:	2.14s
447:	learn:	0.2344196	total:	38.4s	remaining:	2.06s
448:	learn:	0.2344107	total:	38.5s	remaining:	1.97s
449:	learn:	0.2344060	total:	38.5s	remaining:	1.88s
450:	learn:	0.2343996	total:	38.6s	remaining:	1.8s
451:	learn:	0.2343905	total:	38.7s	remaining:	1.71s
452:	learn:	0.2343832	total:	38.7s	remaining:	1.62s
453:	learn:	0.2343750	total:	38.8s	remaining:	1.54s
454:	learn:	0.2343686	total:	38.9s	remaining:	1.45s

```
455: learn: 0.2343599 total: 39s remaining: 1.37s
456: learn: 0.2343547 total: 39s remaining: 1.28s
457: learn: 0.2343462 total: 39.1s remaining: 1.2s
458: learn: 0.2343391 total: 39.2s remaining: 1.11s
459: learn: 0.2343330 total: 39.2s remaining: 1.02s
460: learn: 0.2343266 total: 39.3s remaining: 938ms
461: learn: 0.2343200 total: 39.4s remaining: 853ms
462: learn: 0.2343112 total: 39.5s remaining: 767ms
463: learn: 0.2343075 total: 39.5s remaining: 682ms
464: learn: 0.2342983 total: 39.6s remaining: 596ms
465: learn: 0.2342916 total: 39.7s remaining: 511ms
466: learn: 0.2342870 total: 39.7s remaining: 426ms
467: learn: 0.2342813 total: 39.8s remaining: 340ms
468: learn: 0.2342753 total: 39.9s remaining: 255ms
469: learn: 0.2342665 total: 40s remaining: 170ms
470: learn: 0.2342602 total: 40s remaining: 85ms
471: learn: 0.2342515 total: 40.1s remaining: 0us
```

Model Evaluation

```
from sklearn.metrics import accuracy_score

accuracy_xgb = accuracy_score(y_test, xgb_pred)
accuracy_lgbm = accuracy_score(y_test, lgbm_pred)
accuracy_cat = accuracy_score(y_test, cat_pred)

print(f"XGBoost Accuracy: {accuracy_xgb}")
print(f"LightGBM Accuracy: {accuracy_lgbm}")
print(f"CatBoost Accuracy: {accuracy_cat}")

XGBoost Accuracy: 0.9263601694356105
LightGBM Accuracy: 0.9264723347540941
CatBoost Accuracy: 0.9264723347540941
```

the result shows that both lightGBM and CatBoost has the high accuracy on testing dataset

```
train_pred_xgb = xgb_model.predict(X_train)
train_pred_lgbm = lgbm_model.predict(X_train)
train_pred_cat = cat_model.predict(X_train)

accuracytrain_xgb = accuracy_score(y_train, train_pred_xgb)
accuracytrain_lgbm = accuracy_score(y_train, train_pred_lgbm)
accuracytrain_cat = accuracy_score(y_train, train_pred_cat)

print(f"XGBoost Accuracy: {accuracytrain_xgb}")
print(f"LightGBM Accuracy: {accuracytrain_lgbm}")
print(f"CatBoost Accuracy: {accuracytrain_cat}")
```



```
XGBoost Accuracy: 0.9286825168813836
LightGBM Accuracy: 0.928727760120347
CatBoost Accuracy: 0.9287221047154766
```

the result shows that both lightGBM has the high accuracy on training dataset

```
# accuracy difference between training and testing

print(f"XGBoost Accuracy Difference: {accuracy_xgb -
accuracytrain_xgb}")
print(f"LightGBM Accuracy Difference: {accuracy_lgbm -
accuracytrain_lgbm}")
print(f"CatBoost Accuracy Difference: {accuracy_cat -
accuracytrain_cat}")

XGBoost Accuracy Difference: -0.0023223474457730875
LightGBM Accuracy Difference: -0.0022554253662528945
CatBoost Accuracy Difference: -0.002249769961382486
```

the accuracy results shows that our model is not overfitting at it shows a close gap difference between the training and testing dataset accuracy

the difference shows that catboost is the most closer gap to its training and testing accuracy

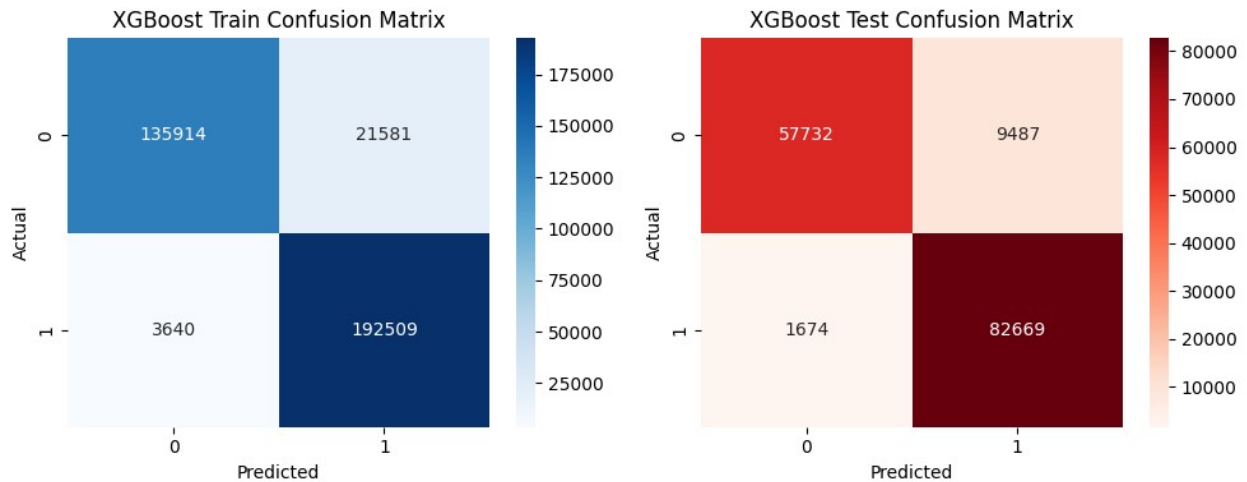
```
from sklearn.metrics import confusion_matrix

fig, ax = plt.subplots(1, 2, figsize=(10, 4))

cm_train = confusion_matrix(y_train, train_pred_xgb)
sns.heatmap(cm_train, annot=True, cmap='Blues', ax=ax[0], fmt='d')
ax[0].set_title('XGBoost Train Confusion Matrix')
ax[0].set_xlabel('Predicted')
ax[0].set_ylabel('Actual')

cm_test = confusion_matrix(y_test, xgb_pred)
sns.heatmap(cm_test, annot=True, cmap='Reds', ax=ax[1], fmt='d')
ax[1].set_title('XGBoost Test Confusion Matrix')
ax[1].set_xlabel('Predicted')
ax[1].set_ylabel('Actual')

plt.tight_layout()
```

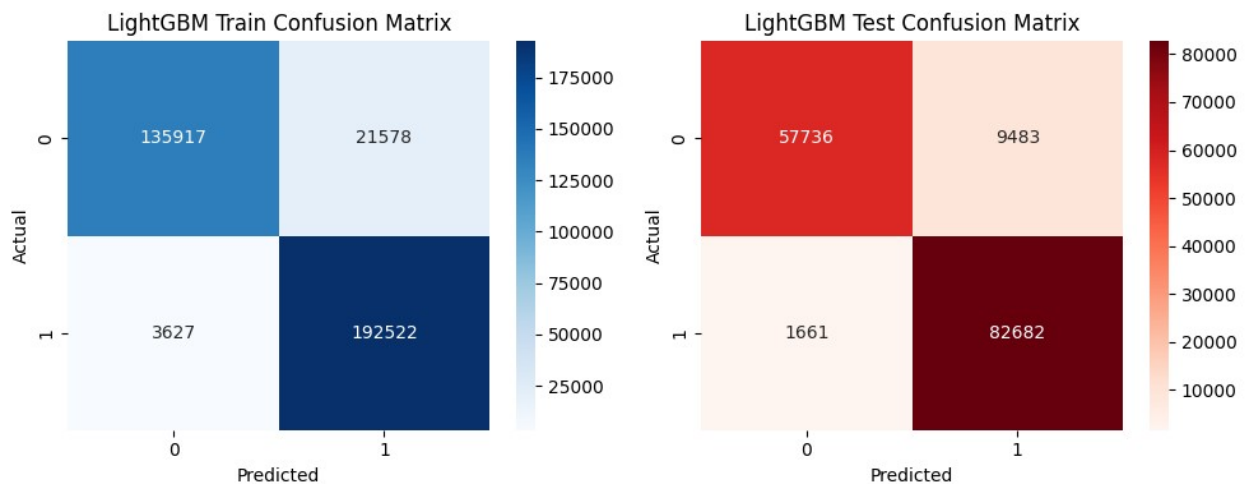


```
fig, ax = plt.subplots(1, 2, figsize=(10, 4))

cm_train = confusion_matrix(y_train, train_pred_lgbm)
sns.heatmap(cm_train, annot=True, cmap='Blues', ax=ax[0], fmt='d')
ax[0].set_title('LightGBM Train Confusion Matrix')
ax[0].set_xlabel('Predicted')
ax[0].set_ylabel('Actual')

cm_test = confusion_matrix(y_test, lgbm_pred)
sns.heatmap(cm_test, annot=True, cmap='Reds', ax=ax[1], fmt='d')
ax[1].set_title('LightGBM Test Confusion Matrix')
ax[1].set_xlabel('Predicted')
ax[1].set_ylabel('Actual')

plt.tight_layout()
```



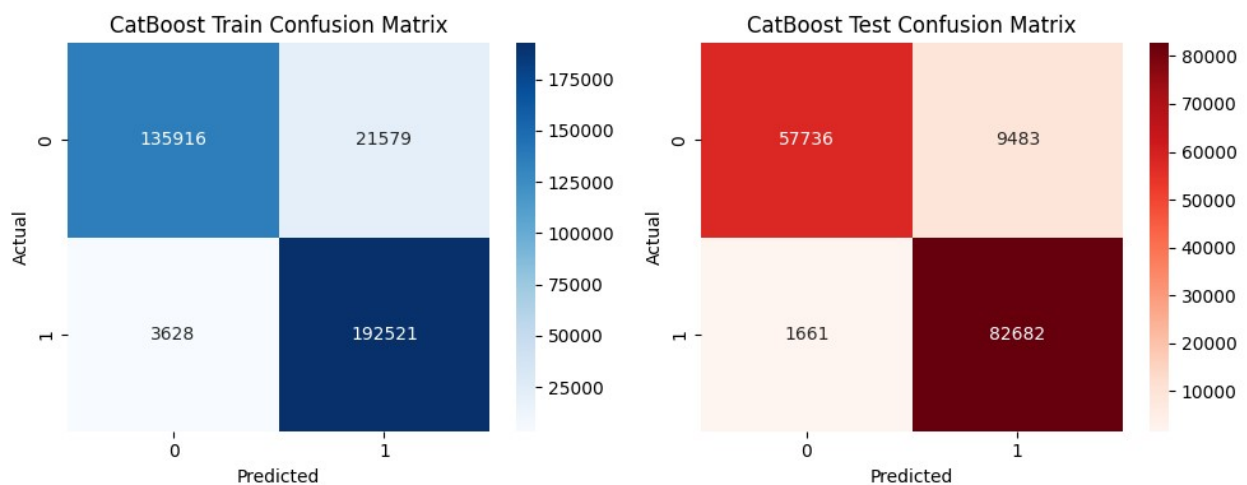
```
fig, ax = plt.subplots(1, 2, figsize=(10, 4))

cm_train = confusion_matrix(y_train, train_pred_cat)
```

```
sns.heatmap(cm_train, annot=True, cmap='Blues', ax=ax[0], fmt='d')
ax[0].set_title('CatBoost Train Confusion Matrix')
ax[0].set_xlabel('Predicted')
ax[0].set_ylabel('Actual')
```

```
cm_test = confusion_matrix(y_test, cat_pred)
sns.heatmap(cm_test, annot=True, cmap='Reds', ax=ax[1], fmt='d')
ax[1].set_title('CatBoost Test Confusion Matrix')
ax[1].set_xlabel('Predicted')
ax[1].set_ylabel('Actual')
```

```
plt.tight_layout()
```



the confusion matrix shows the results of the three models between training and testing dataset, LightGBM is has more correct prediction on people who churned on the training dataset compared to other models, but both CatBoost and LightGBM are the most accurate when it comes in predicting customers who did not churned. on the other hand, LightGBM still is the more accurate in predicting churned customers in training dataset, and again both CatBoost and LightGBM has the same counts of correct predicted churned customers

```
# making a classification report on the 3 models
```

```
from sklearn.metrics import classification_report
```

```
print("XGBoost Classification Report:")
print(classification_report(y_test, xgb_pred))
```

```
print("\nLightGBM Classification Report:")
print(classification_report(y_test, lgbm_pred))
```

```
print("\nCatBoost Classification Report:")
print(classification_report(y_test, cat_pred))
```

XGBoost Classification Report:

	precision	recall	f1-score	support
0	0.97	0.86	0.91	67219
1	0.90	0.98	0.94	84343
accuracy			0.93	151562
macro avg	0.93	0.92	0.92	151562
weighted avg	0.93	0.93	0.93	151562

LightGBM Classification Report:

	precision	recall	f1-score	support
0	0.97	0.86	0.91	67219
1	0.90	0.98	0.94	84343
accuracy			0.93	151562
macro avg	0.93	0.92	0.92	151562
weighted avg	0.93	0.93	0.93	151562

CatBoost Classification Report:

	precision	recall	f1-score	support
0	0.97	0.86	0.91	67219
1	0.90	0.98	0.94	84343
accuracy			0.93	151562
macro avg	0.93	0.92	0.92	151562
weighted avg	0.93	0.93	0.93	151562

plotting the roc auc curve of the 3 models

```
from sklearn.metrics import roc_curve, auc
```

```
fig, ax = plt.subplots(1, 3, figsize=(15, 4))
```

```
fpr_xgb, tpr_xgb, _ = roc_curve(y_test,
xgb_model.predict_proba(X_test)[: , 1])
roc_auc_xgb = auc(fpr_xgb, tpr_xgb)
fpr_xgb_train, tpr_xgb_train, _ = roc_curve(y_train,
xgb_model.predict_proba(X_train)[: , 1])
roc_auc_xgb_train = auc(fpr_xgb_train, tpr_xgb_train)
ax[0].plot(fpr_xgb, tpr_xgb, label=f'XGBoost (AUC =
{round(roc_auc_xgb,4)})')
ax[0].plot(fpr_xgb_train, tpr_xgb_train, label=f'XGBoost Train (AUC =
{round(roc_auc_xgb_train,4)})')
ax[0].set_title('ROC Curve XGB')
ax[0].set_xlabel('False Positive Rate')
```

```

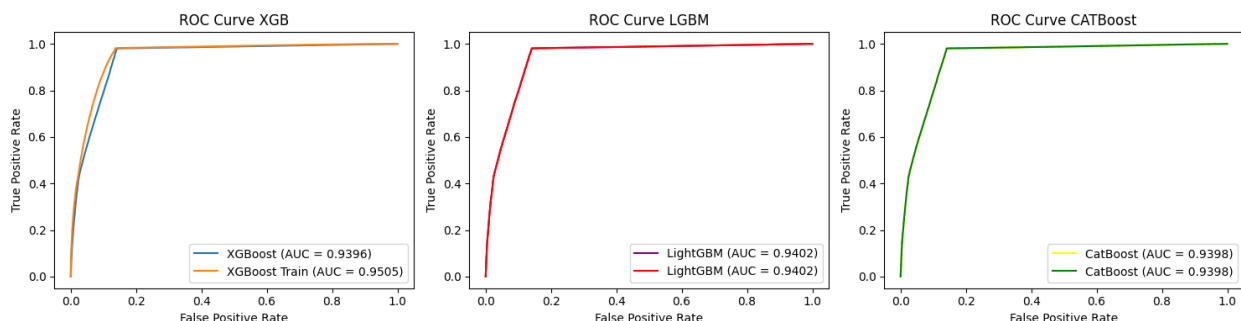
ax[0].set_ylabel('True Positive Rate')
ax[0].legend()

fpr_lgbm, tpr_lgbm, _ = roc_curve(y_test,
lgbm_model.predict_proba(X_test)[: , 1])
roc_auc_lgbm = auc(fpr_lgbm, tpr_lgbm)
fpr_lgbm_train, tpr_lgbm_train, _ = roc_curve(y_train,
lgbm_model.predict_proba(X_train)[: , 1])
roc_auc_lgbm_train = auc(fpr_lgbm_train, tpr_lgbm_train)
ax[1].plot(fpr_lgbm, tpr_lgbm, label=f'LightGBM (AUC =
{round(roc_auc_lgbm,4)})',color='purple')
ax[1].plot(fpr_lgbm, tpr_lgbm, label=f'LightGBM (AUC =
{round(roc_auc_lgbm,4)})', color='red')
ax[1].set_title('ROC Curve LGBM')
ax[1].set_xlabel('False Positive Rate')
ax[1].set_ylabel('True Positive Rate')
ax[1].legend()

fpr_cat, tpr_cat, _ = roc_curve(y_test,
cat_model.predict_proba(X_test)[: , 1])
roc_auc_cat = auc(fpr_cat, tpr_cat)
fpr_cat_train, tpr_cat_train, _ = roc_curve(y_train,
cat_model.predict_proba(X_train)[: , 1])
roc_auc_cat_train = auc(fpr_cat_train, tpr_cat_train)
ax[2].plot(fpr_cat, tpr_cat, label=f'CatBoost (AUC =
{round(roc_auc_cat,4)})', color='yellow')
ax[2].plot(fpr_cat, tpr_cat, label=f'CatBoost (AUC =
{round(roc_auc_cat,4)})', color='green')
ax[2].set_title('ROC Curve CATBoost')
ax[2].set_xlabel('False Positive Rate')
ax[2].set_ylabel('True Positive Rate')
ax[2].legend()

plt.tight_layout()

```



Three gradient boosting models were trained and evaluated to predict customer churn: XGBoost, LightGBM, and CatBoost. All three models were tuned using Optuna with 5-Fold Stratified Cross Validation. Results show that LightGBM the highest test accuracy, with

CatBoost showing the smallest gap between training and testing accuracy it indicates it is the most stable and generalizable model among the three. All three models showed no significant signs of overfitting, as the difference between training and testing accuracy remained close across all models.

Saving the model

```
import joblib
import os
os.makedirs('/content/drive/MyDrive/projects/models/churnproject',
            exist_ok=True)

joblib.dump(xgb_model,
            '/content/drive/MyDrive/projects/models/churnproject/xgboost_model.pkl')
joblib.dump(lgbm_model,
            '/content/drive/MyDrive/projects/models/churnproject/lightgbm_model.pkl')
joblib.dump(cat_model,
            '/content/drive/MyDrive/projects/models/churnproject/catboost_model.pkl')

print("All models saved")
```

All models saved